

Announcements

- Midterm Grades Released
 - 75% average for Section C
- HW3 ongoing

Regularized Linear Regression

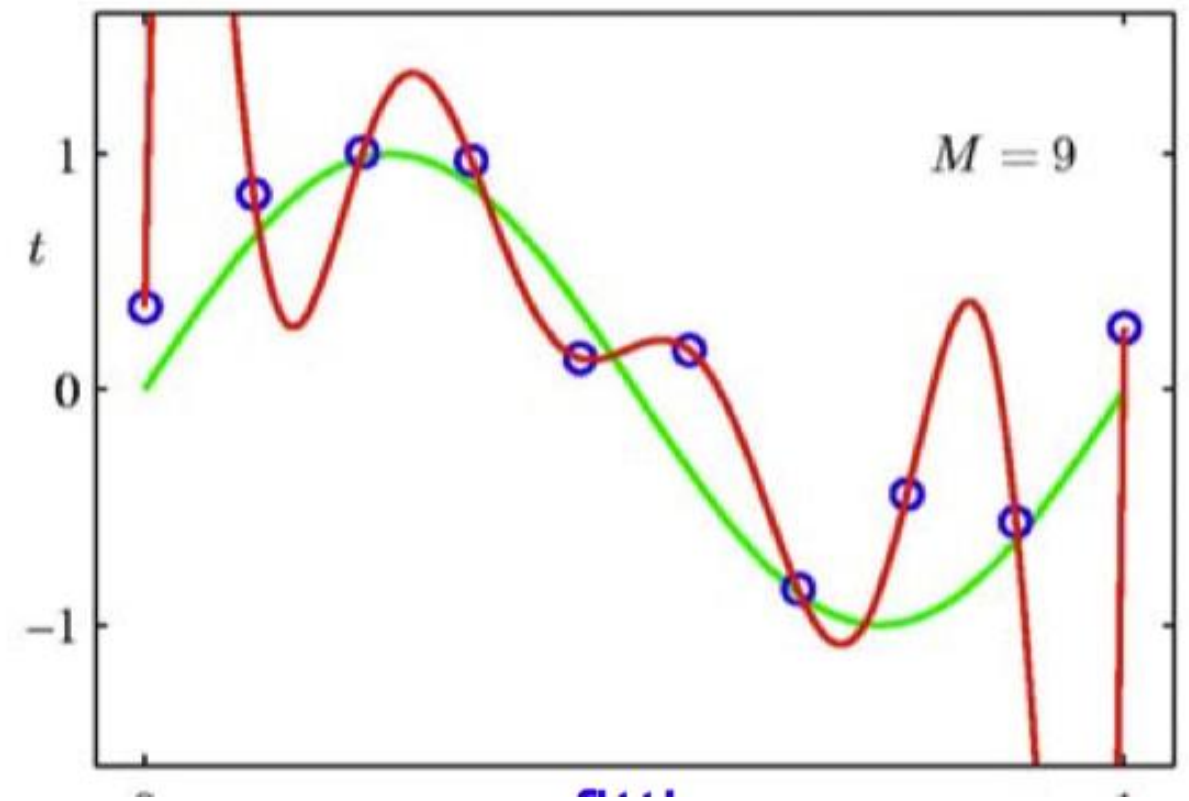
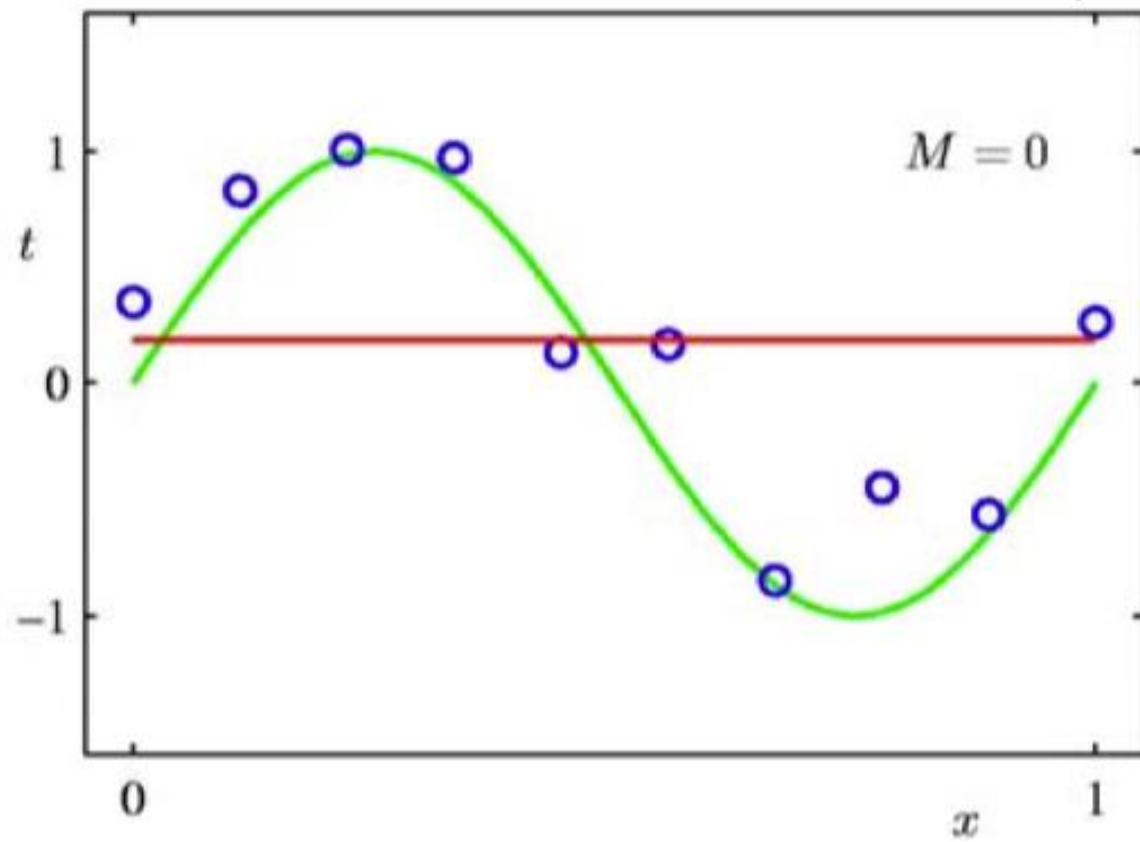
Nimisha Roy
Georgia Tech

Outline

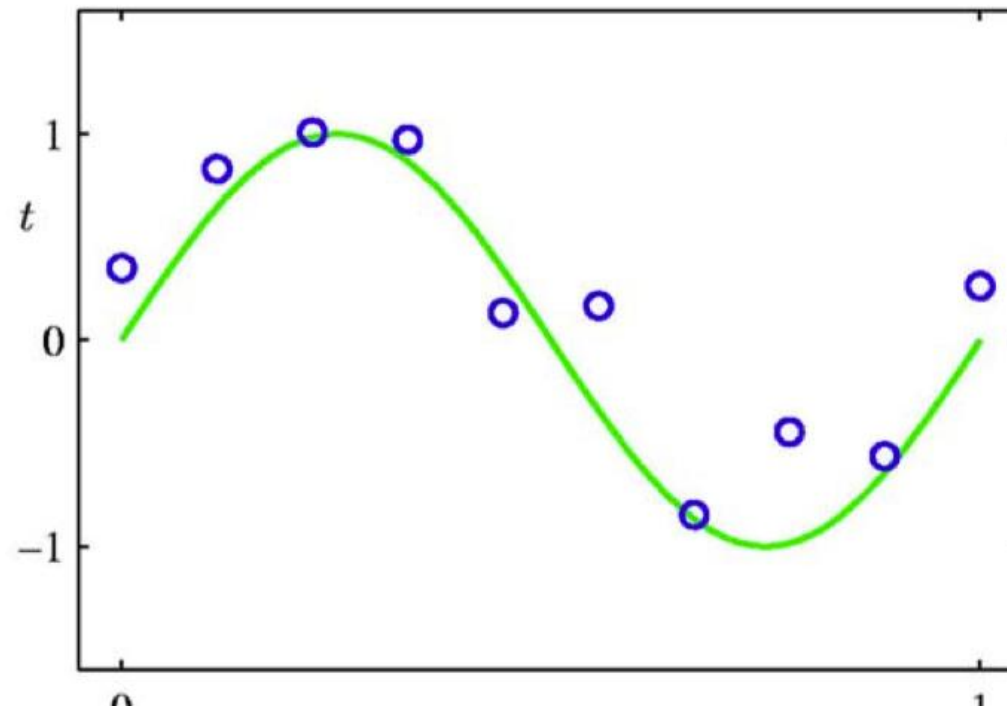
- Overfitting and regularized learning
- Ridge regression
- Lasso regression
- Determining regularization strength



Recap: Overfitting vs Underfitting



Regression: Why is overfitting more common?



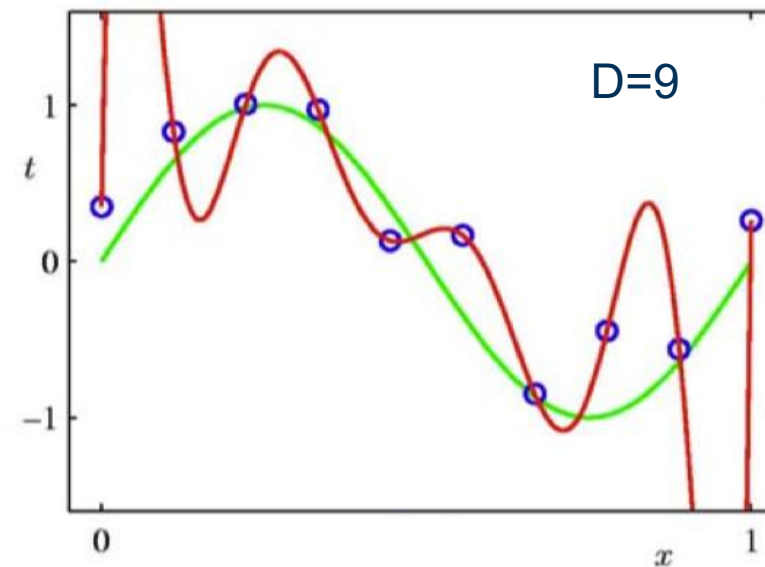
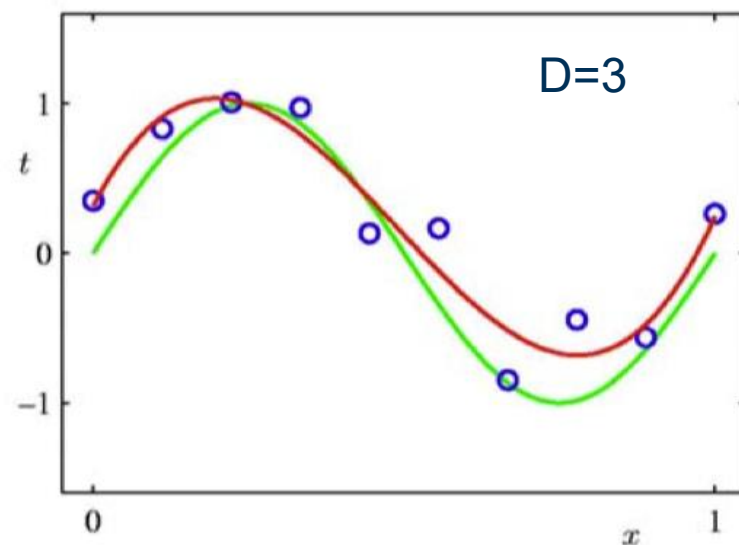
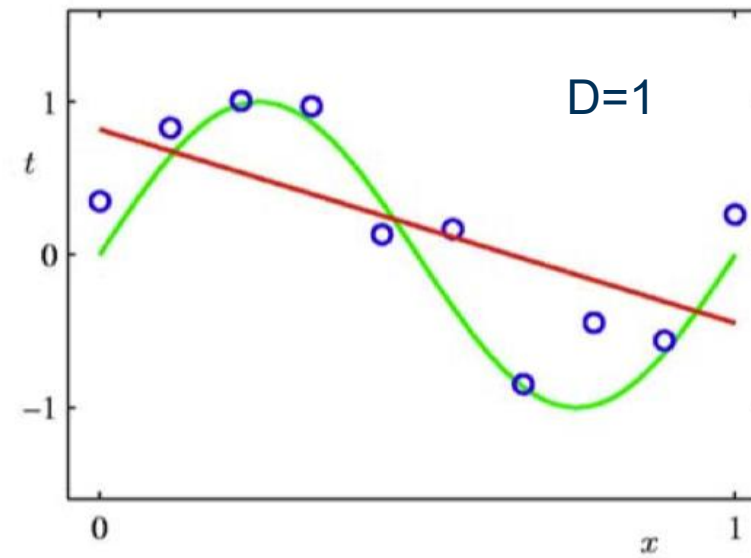
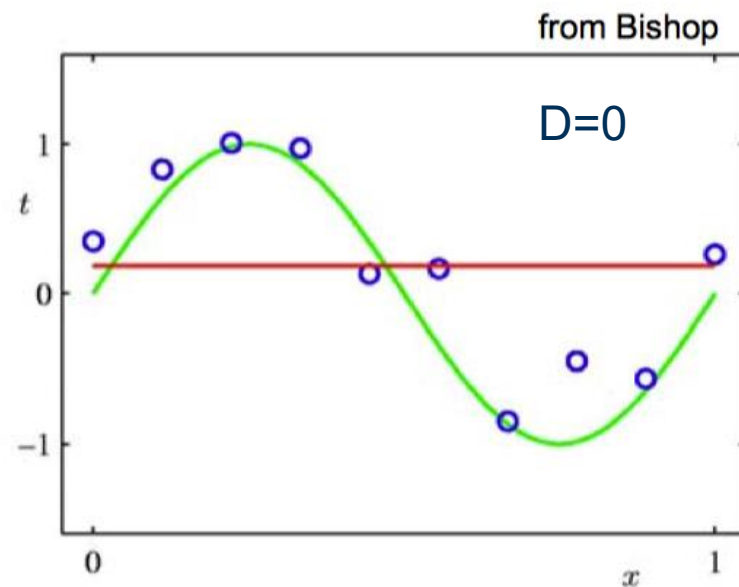
- Want to fit a polynomial regression model

$$y = \theta_0 + \theta_1 x + \theta_2 x^2 + \dots + \theta_d x^d + \epsilon$$

- $z = \{1, x, x^2, \dots, x^d\} \in R^d$ and $\theta = (\theta_0, \theta_1, \theta_2, \dots, \theta_d)^T$

$$y = z\theta$$

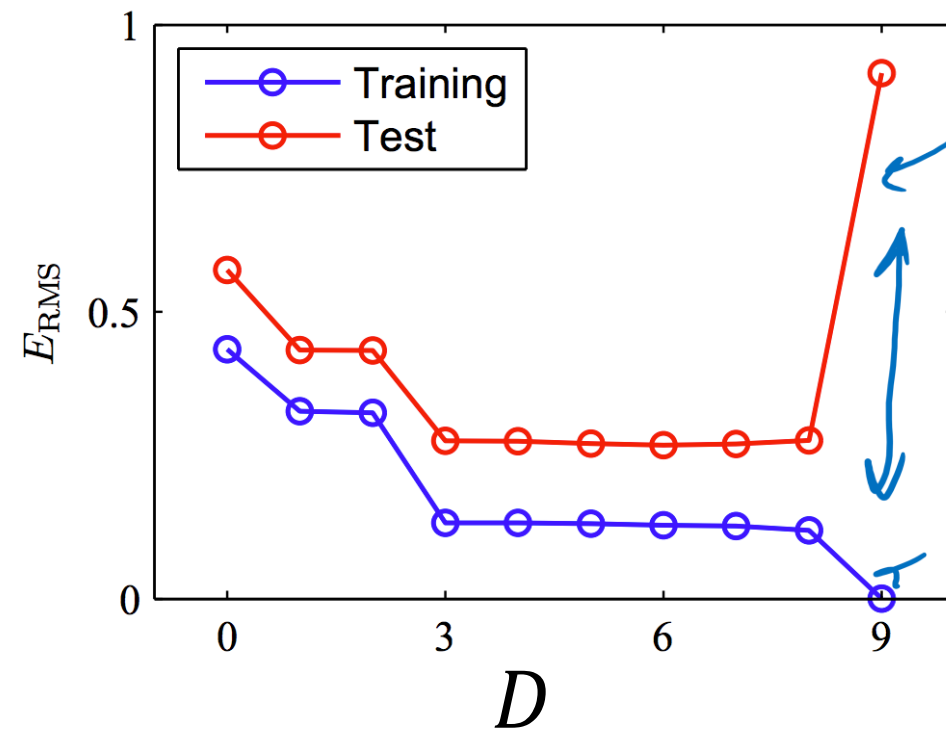
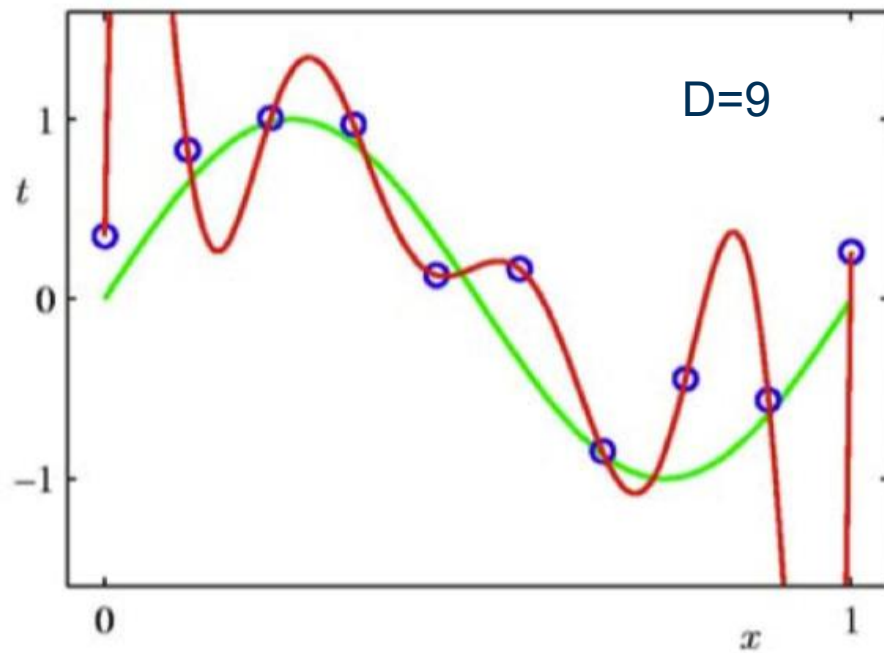
Which One is Better?



- Can we increase the maximal polynomial degree to very large, such that the curve passes through all training points?

No, this can lead to **overfitting**!

The Overfitting Problem

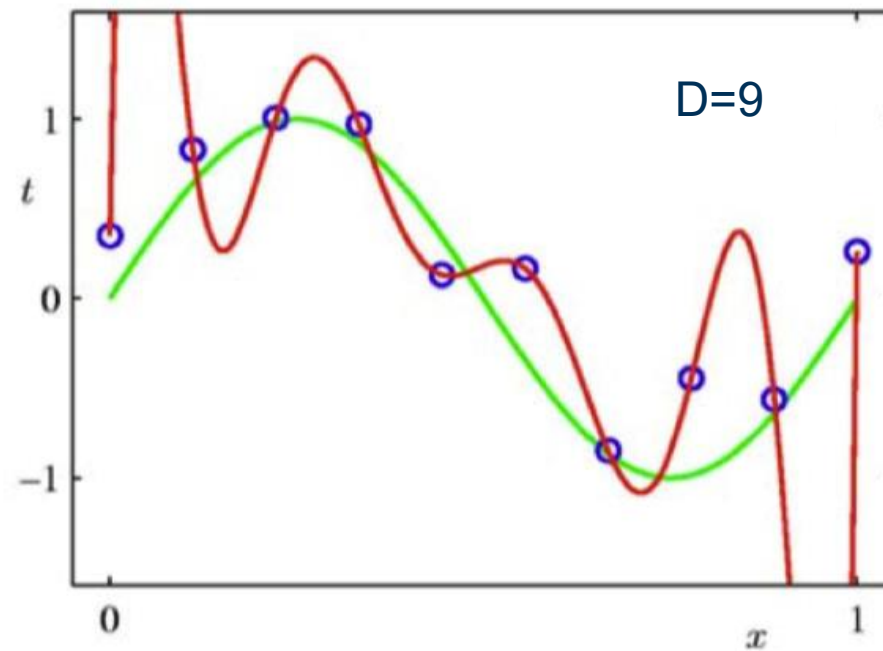


- The training error is very low, but the error on test set is large.
- The model captures not only patterns but also noisy nuisances in the training data.

Root cause of overfitting?

- I want to convert 1 feature in x space to d features in z space

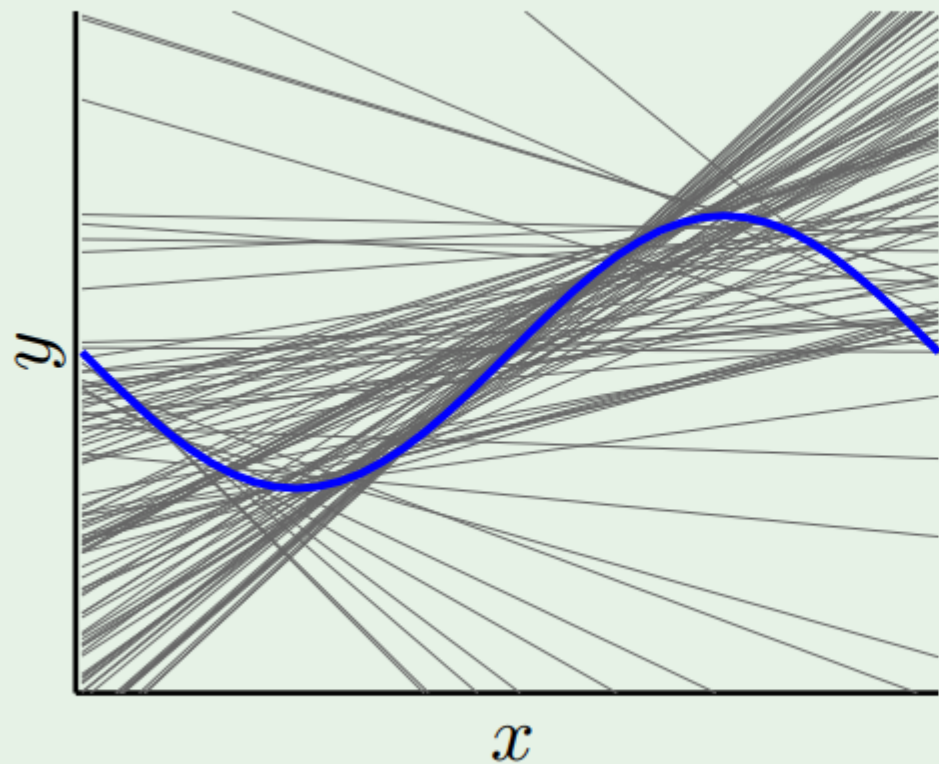
The Overfitting Problem



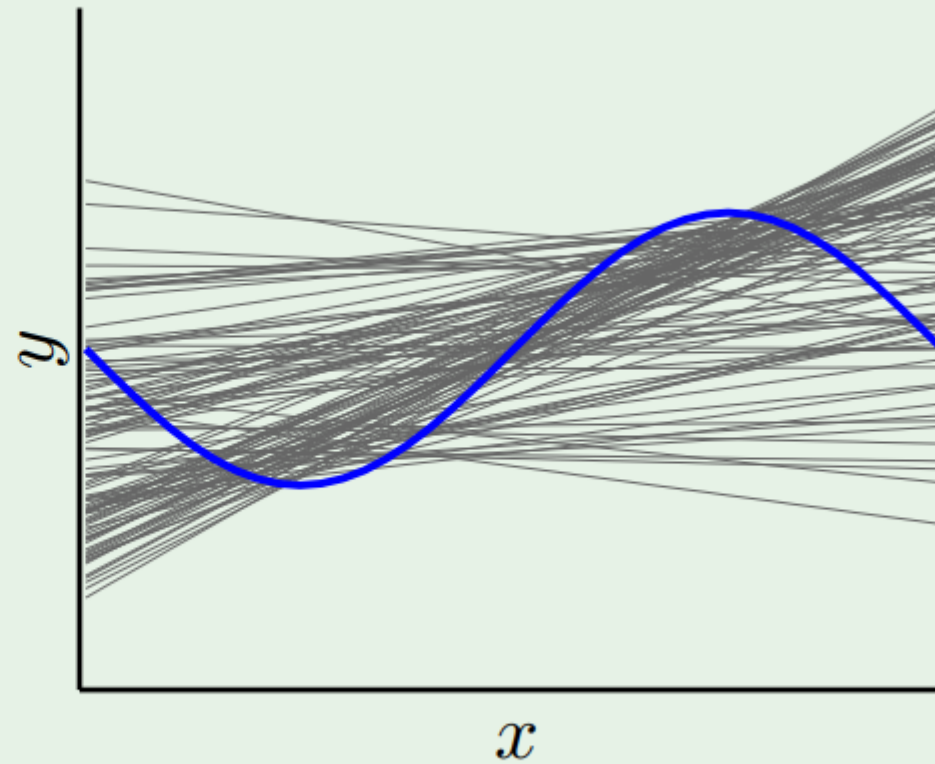
- In regression, overfitting is often associated with large Weights (severe oscillation)
- How can we address overfitting?

Regularization

(smart way to cure overfitting disease)



without regularization



with regularization

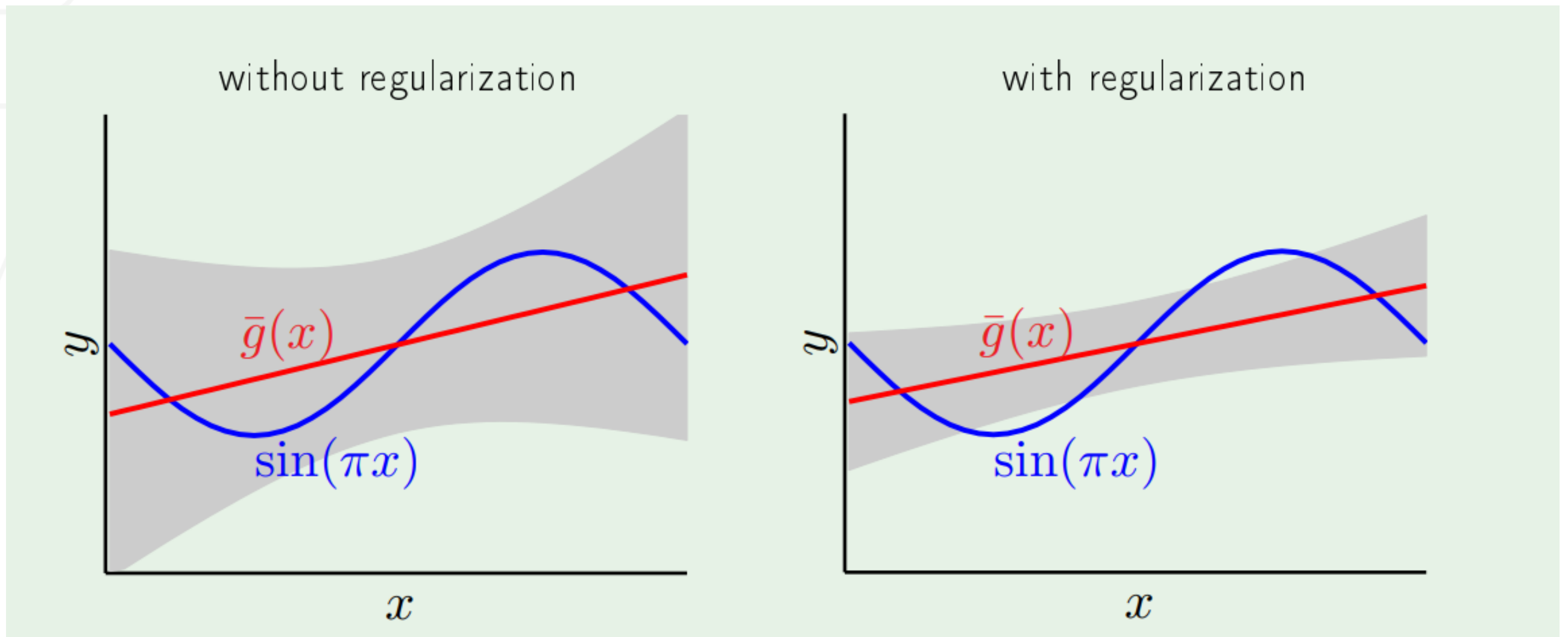
Put a brake on fitting



Fit a linear line on sinusoidal with just two data points

Who is the winner?

$\bar{g}(x)$: average over all lines



bias=0.21; var=1.69

bias=0.23; var=0.33

So, what is regularization in general?

Regularization adds a penalty term to the error or objective function:

- The penalty discourages large weights → the model becomes *simpler*.
- A simpler model can't bend perfectly to fit all training data → **bias increases slightly**.
- But since it's not so sensitive to the quirks of any one dataset → **variance decreases significantly**.
- That tradeoff leads to better **generalization**.

Regularization	Bias	Variance	Total Error	Stability
✗ Without	Low	High	High	Overfits
✓ With	Slightly higher	Much lower	Lower	Generalizes better

Polynomial Model

Want to fit a polynomial regression model

$$y = \theta_0 + \theta_1 x + \theta_2 x^2 + \cdots + \theta_d x^d + \epsilon$$

Let's rewrite it as:

$$y = \theta_0 + \theta_1 z_1 + \theta_2 z_2 + \cdots + \theta_d z_d + \epsilon = \mathbf{z}\boldsymbol{\theta}$$

Regularizing is just constraining the weights (θ)

For example: let's do a **hard** constraining

$$y = \theta_0 + \theta_1 z_1 + \theta_2 z_2 + \cdots + \theta_d z_d$$

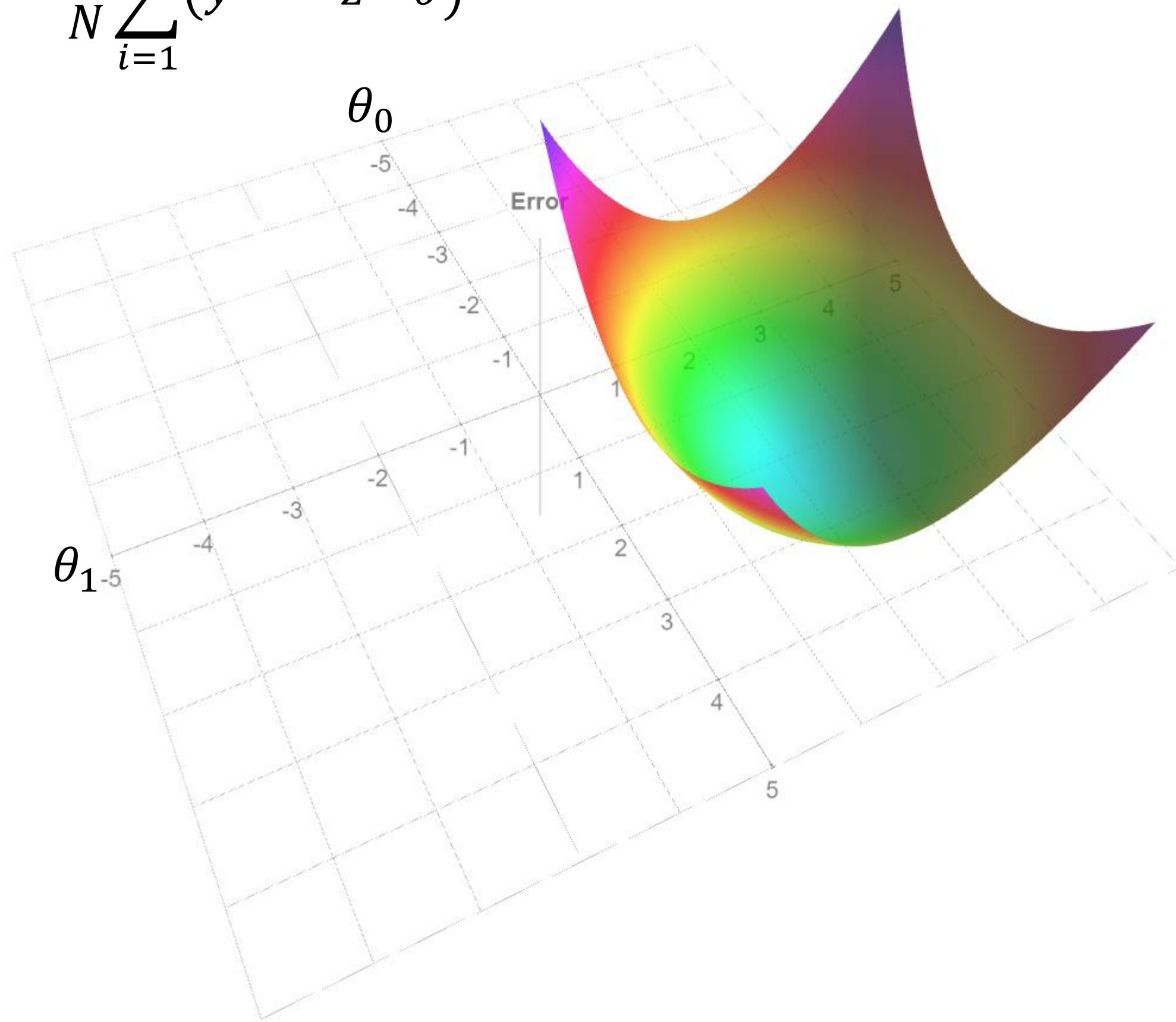
subject to

$$\theta_d = 0 \text{ for } d > 2$$

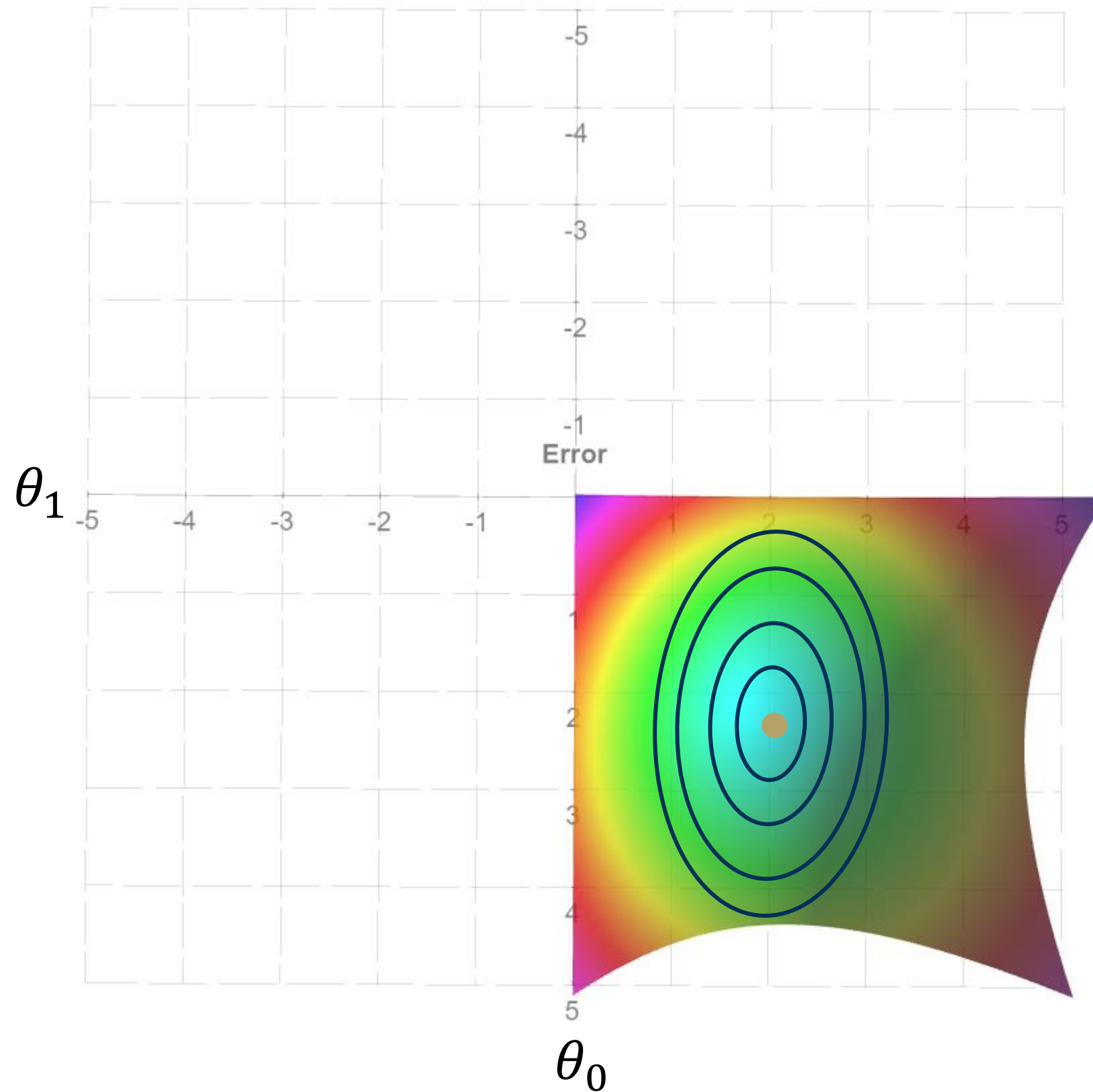


$$y = \theta_0 + \theta_1 z_1 + \theta_2 z_2 + 0 + \cdots + 0$$

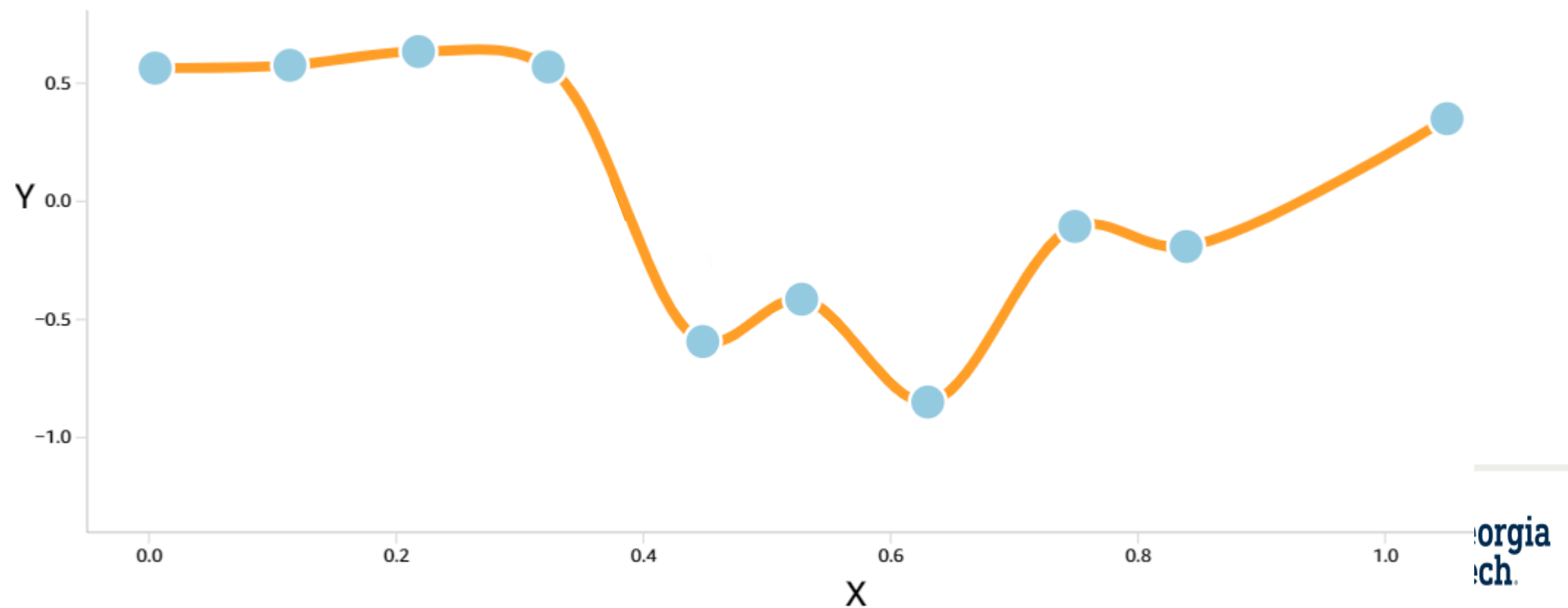
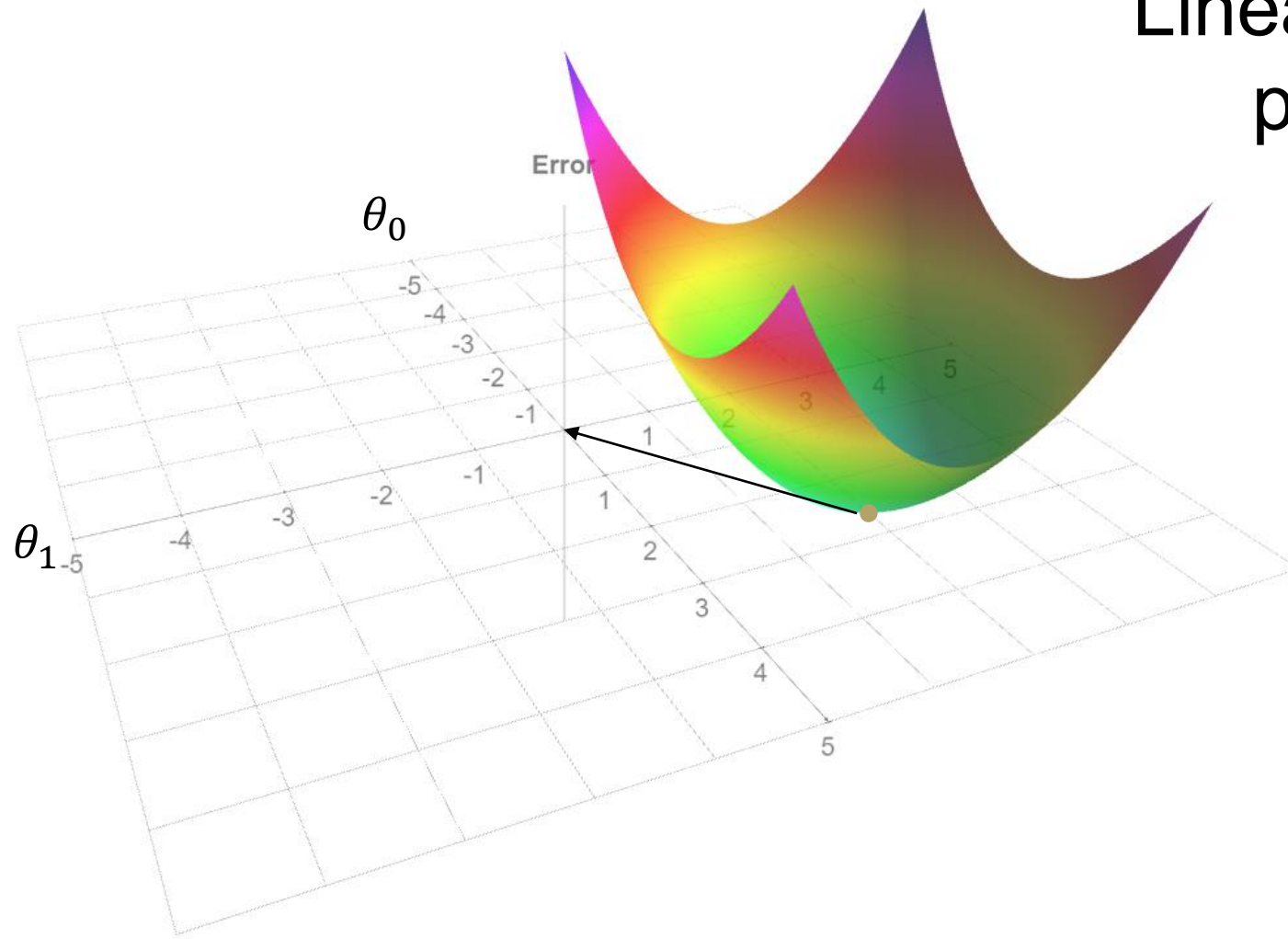
$$E(\theta) = \frac{1}{N} \sum_{i=1}^n (y^{\{i\}} - z^{\{i\}} \theta)^2$$

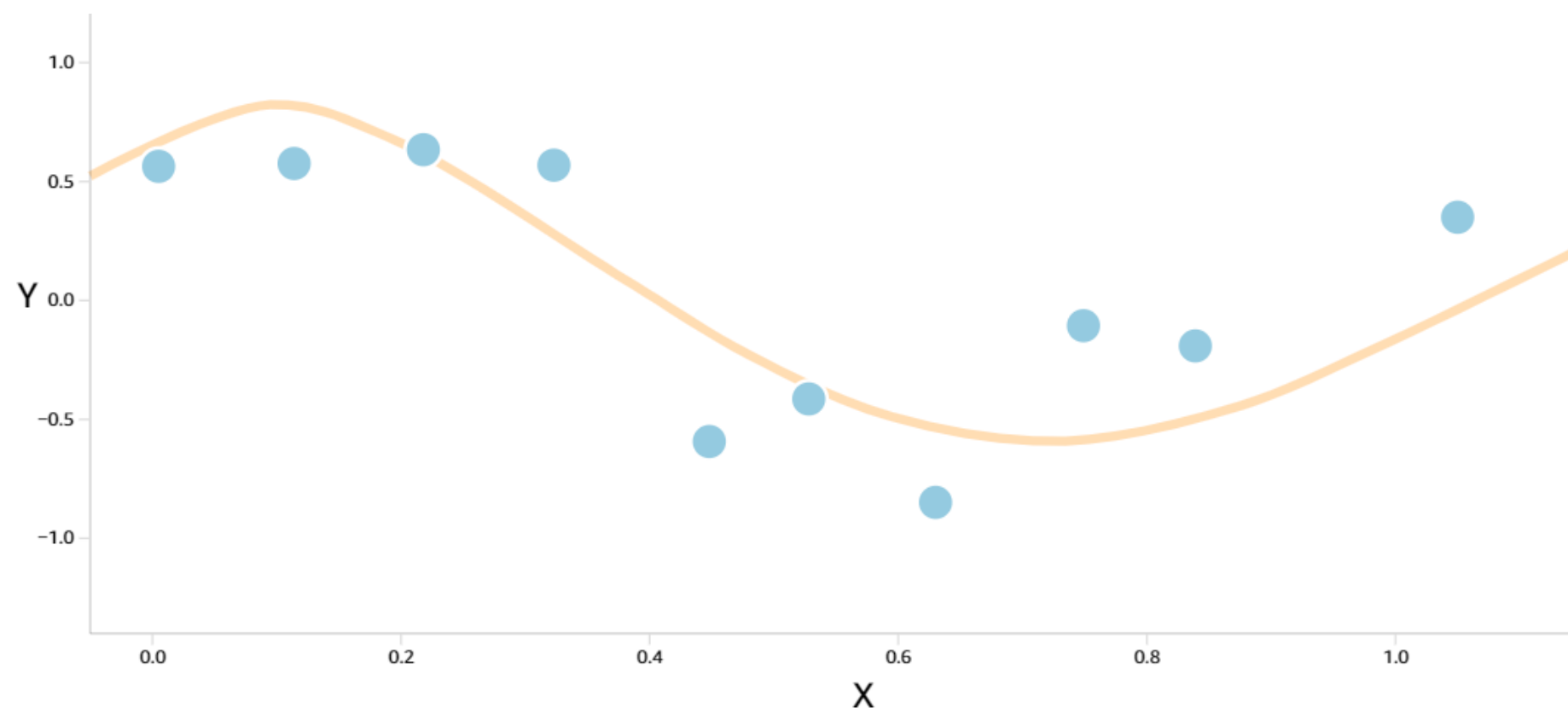
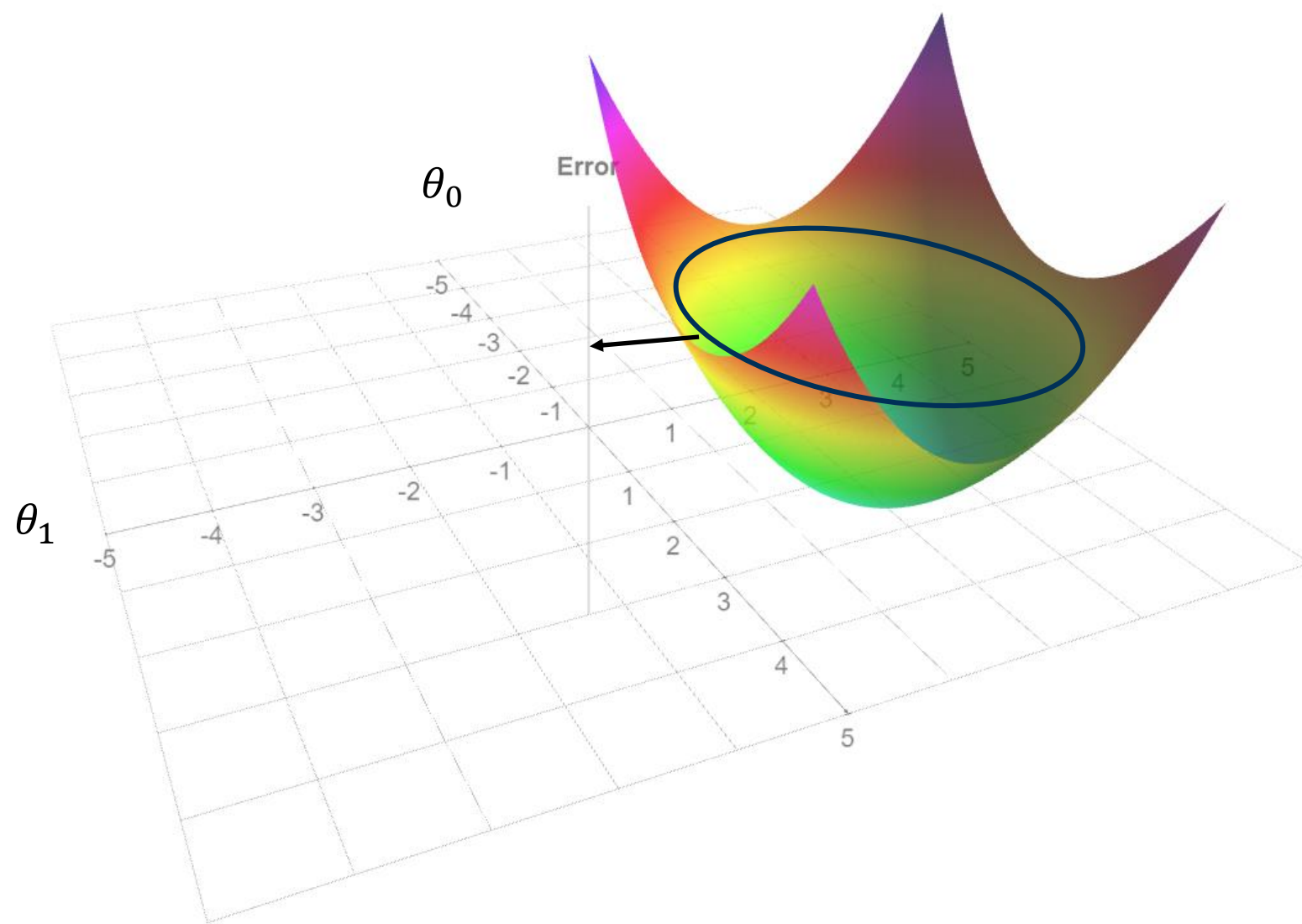


Project the same graph on x-y using contour plot



Linear regression with a very high polynomial degree solution



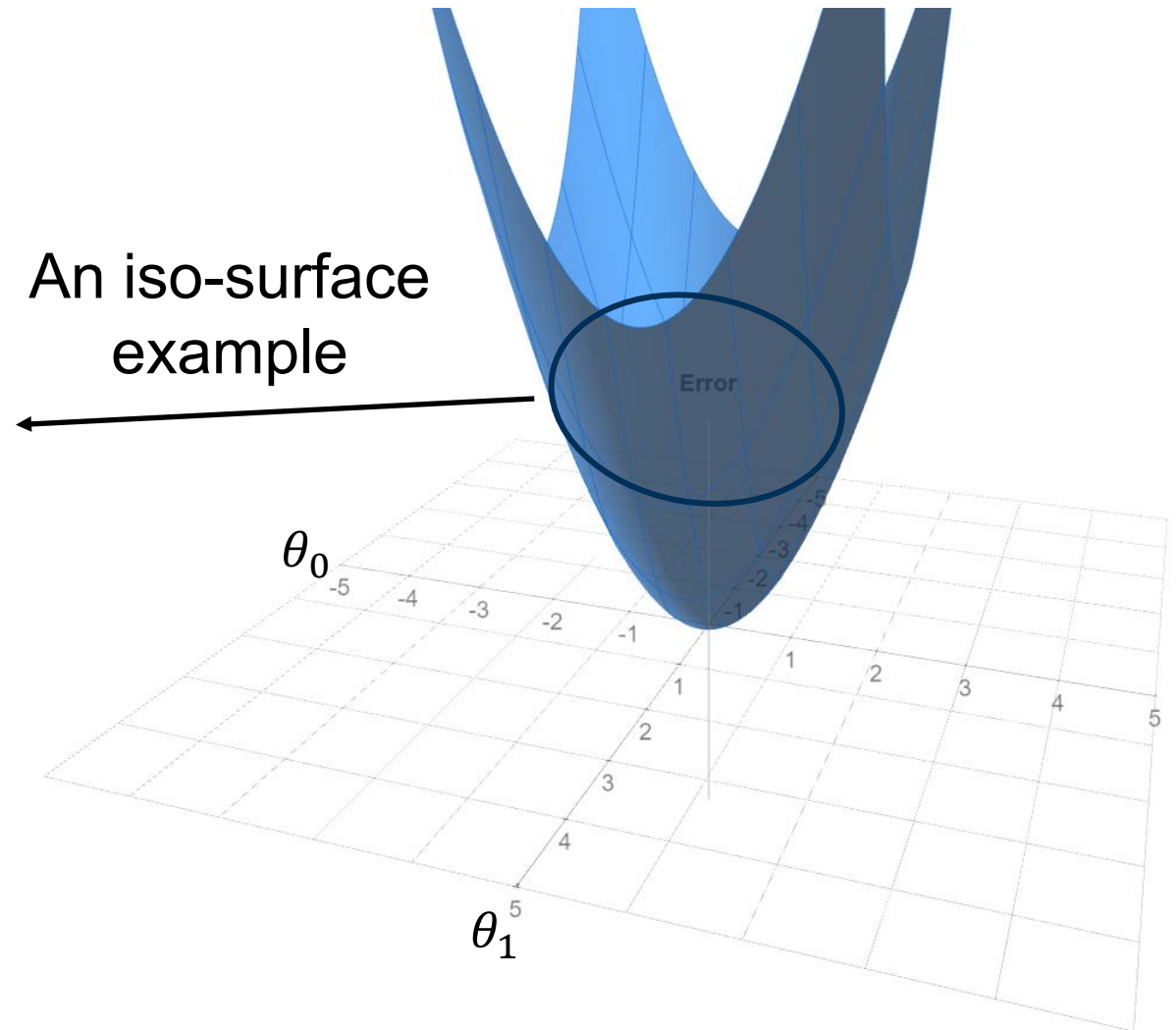


How can we get an optimal solution with a positive error for a model that overfits?

We need to introduce a constraint

$$g(\theta) = \theta_0^2 + \theta_1^2 = \theta^T \theta = C$$

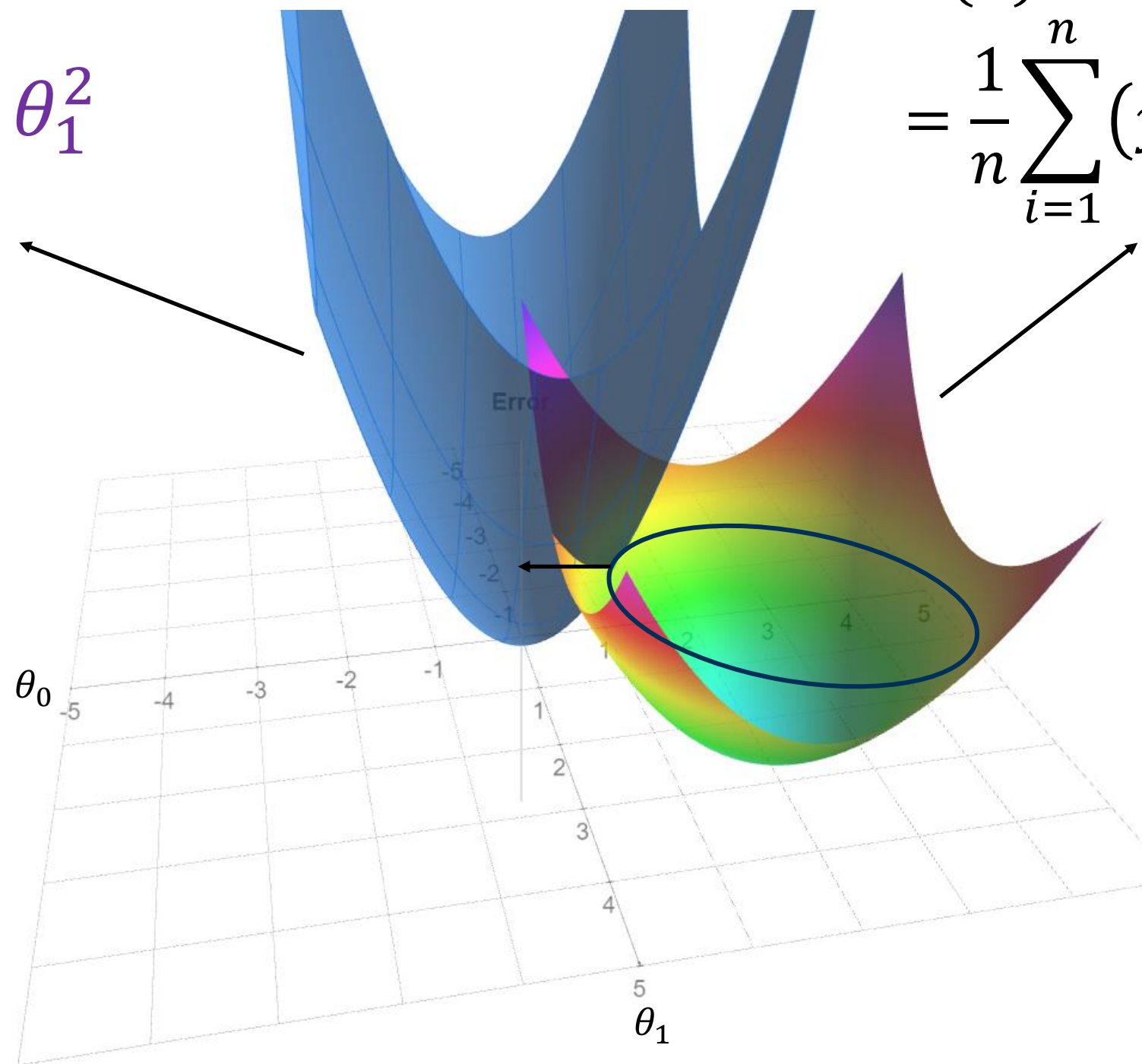
An iso-surface example



Error function together with a new introduced constraint

$$g(\theta) = \theta_0^2 + \theta_1^2 \\ = \theta^T \theta$$

$$E(\theta) \\ = \frac{1}{n} \sum_{i=1}^n (y^{\{i\}} - z^{\{i\}} \theta)^2$$



Let's define the Lagrange function

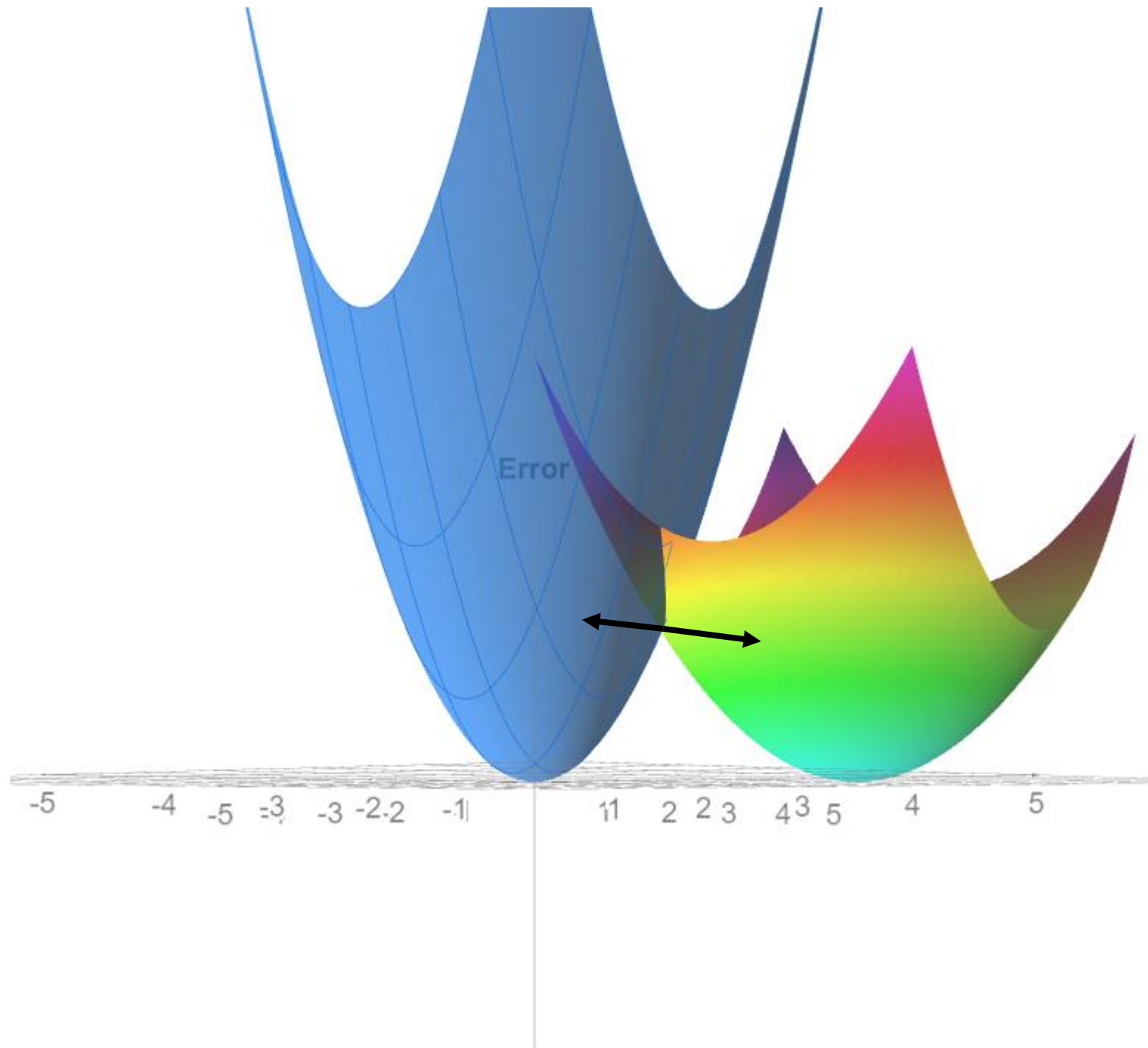
$$L(\theta, \lambda) = E(\theta) + \lambda g(\theta)$$

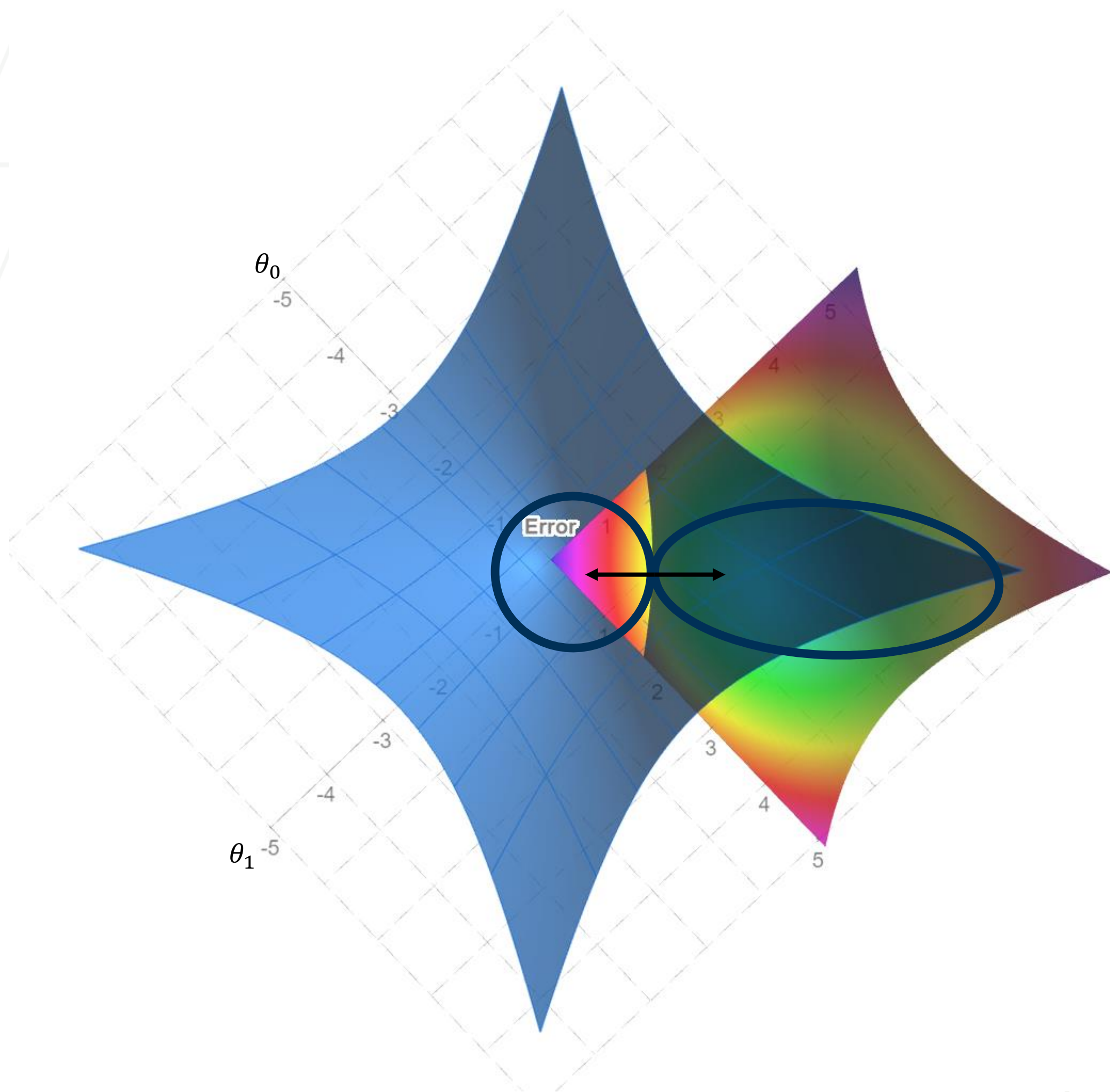
$$L(\theta, \lambda) = E(\theta) + \lambda \theta^T \theta - \lambda C$$

$$\nabla L(\theta, \lambda) = 0 \quad \nabla [E(\theta) + \lambda \theta^T \theta - \lambda C] = 0$$

$$\nabla [E(\theta)] + \lambda \nabla [\theta^T \theta] = 0$$

How to enforce the gradient of Lagrange function to be zero





$$L = E(\theta) + \lambda \theta^T \theta - \lambda C$$

$$\frac{\partial L}{\partial \theta} = 0 \quad \& \quad \frac{\partial L}{\partial \lambda} = 0 \quad \text{gives implicit solution.}$$

Hence, λ is considered a hyperparameter

$$\tilde{E}(\theta) = E(\theta) + \underbrace{\lambda \theta^T \theta}_{\text{penalty term}}$$

Regularized Error

Let's calculate the gradients

Gradient of constraint $g(\theta)$ $\nabla[\theta^T \theta] = 2\theta$

$$\nabla[E(\theta)] + \lambda \nabla[\theta^T \theta] = 0$$

$$\nabla[E(\theta)] = -\lambda \nabla[\theta^T \theta]$$

$$\nabla E(\theta) = -2\lambda\theta$$

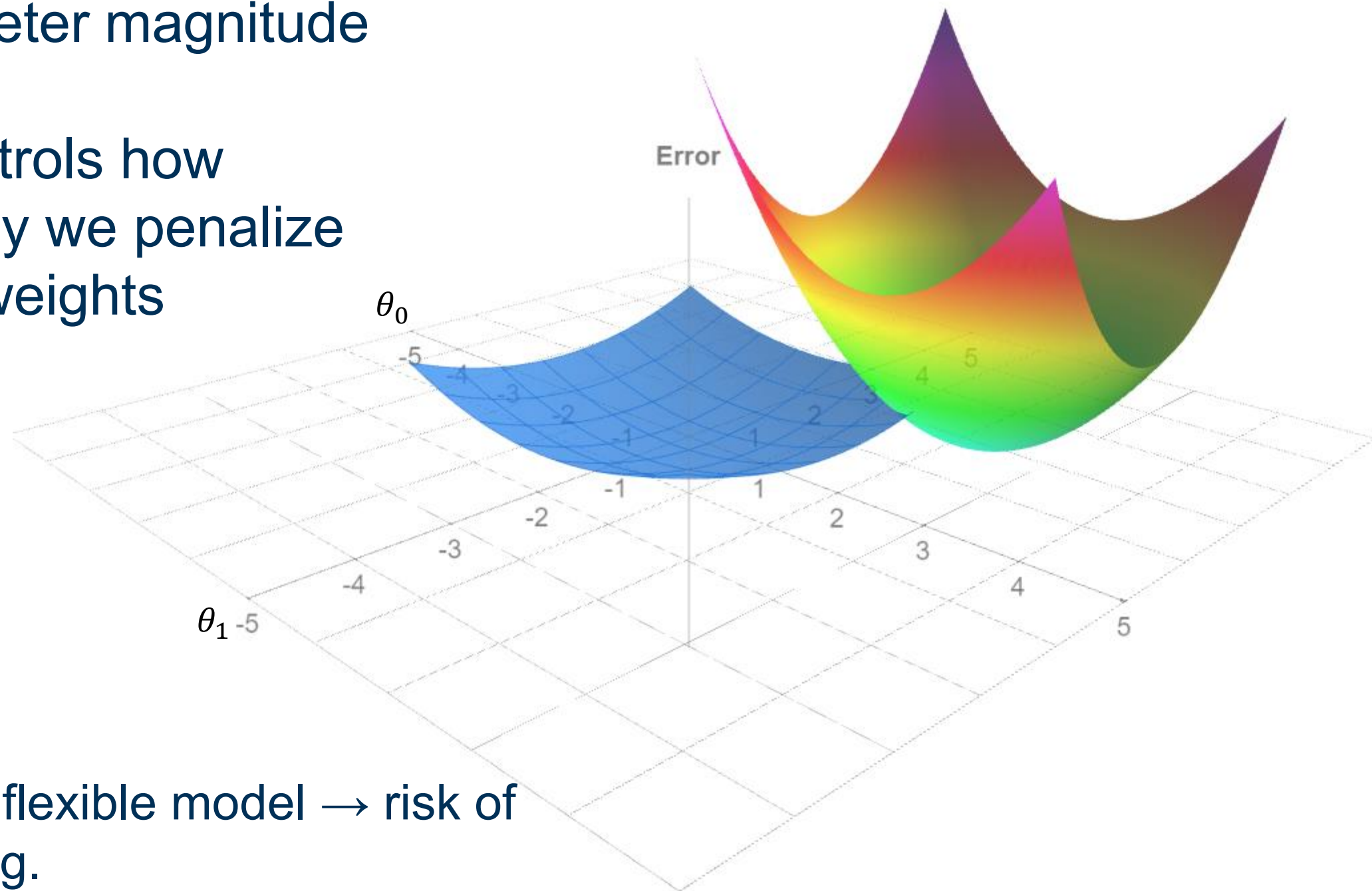
$$\left. \nabla E(\theta) + 2\lambda\theta = 0 \right\} \text{ Let's do integration } E(\theta) + \lambda\theta^T \theta$$

The effect of low Lambda

$\frac{\lambda}{N} \theta^T \theta$: The **regularization penalty** on parameter magnitude

λ : Controls how strongly we penalize large weights

$$E(\theta) + \frac{\lambda}{N} \theta^T \theta$$



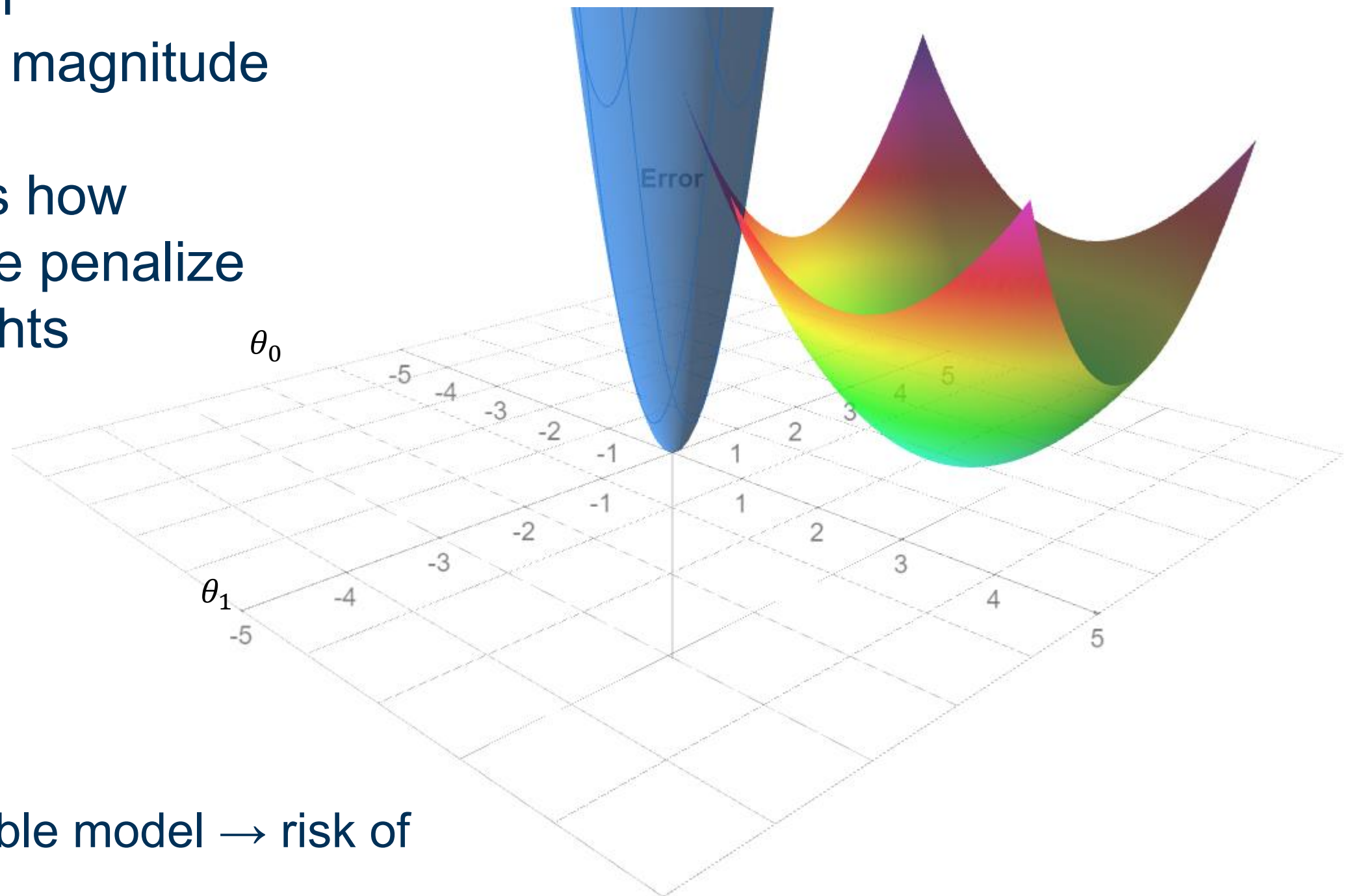
low $\lambda \rightarrow$ flexible model $\rightarrow$ risk of overfitting.

The effect of high Lambda

$\frac{\lambda}{N} \theta^T \theta$: The **regularization penalty** on parameter magnitude

λ : Controls how strongly we penalize large weights

$$E(\theta) + \frac{\lambda}{N} \theta^T \theta$$



high $\lambda \rightarrow$ stable model $\rightarrow$ risk of underfitting.

Regularized Learning

Minimize $E(\theta) + \lambda \theta^T \theta$

Now we know Why this term leads to the regularization of parameters

Regularized Error

$$\tilde{E}(\theta) = \frac{1}{N} \sum_{i=1}^n (y^{\{i\}} - z^{\{i\}} \theta)^2 + \frac{\lambda}{2N} \|\theta\|_2^2$$

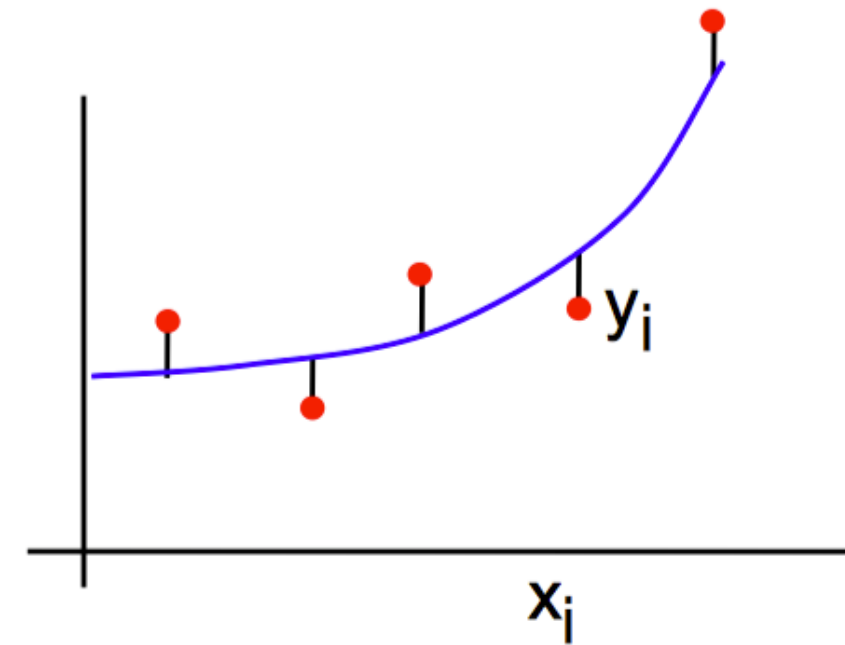
L2 Regularization term

Outline

- Overfitting and regularized learning
- Ridge regression 
- Lasso regression
- Determining regularization strength

Ridge Regression

$$\tilde{E}(\theta) = \frac{1}{N} \sum_{i=1}^n (y^{\{i\}} - z^{\{i\}} \theta)^2 + \frac{\lambda}{2N} \|\theta\|_2^2$$



$$\theta_0 + \theta_1 z_1 + \theta_2 z_2 + \cdots + \theta_d z_d + \epsilon = \mathbf{z} \theta$$

General form

$$\tilde{E}(\theta) = \frac{1}{N} \sum_{i=1}^n (y^{\{i\}} - z^{\{i\}} \theta)^2 + \lambda \|\theta\|_2^2$$

Matrix form

$$\tilde{E}(\theta) = \frac{1}{N} (y - z\theta)^T (y - z\theta) + \lambda \|\theta\|_2^2$$

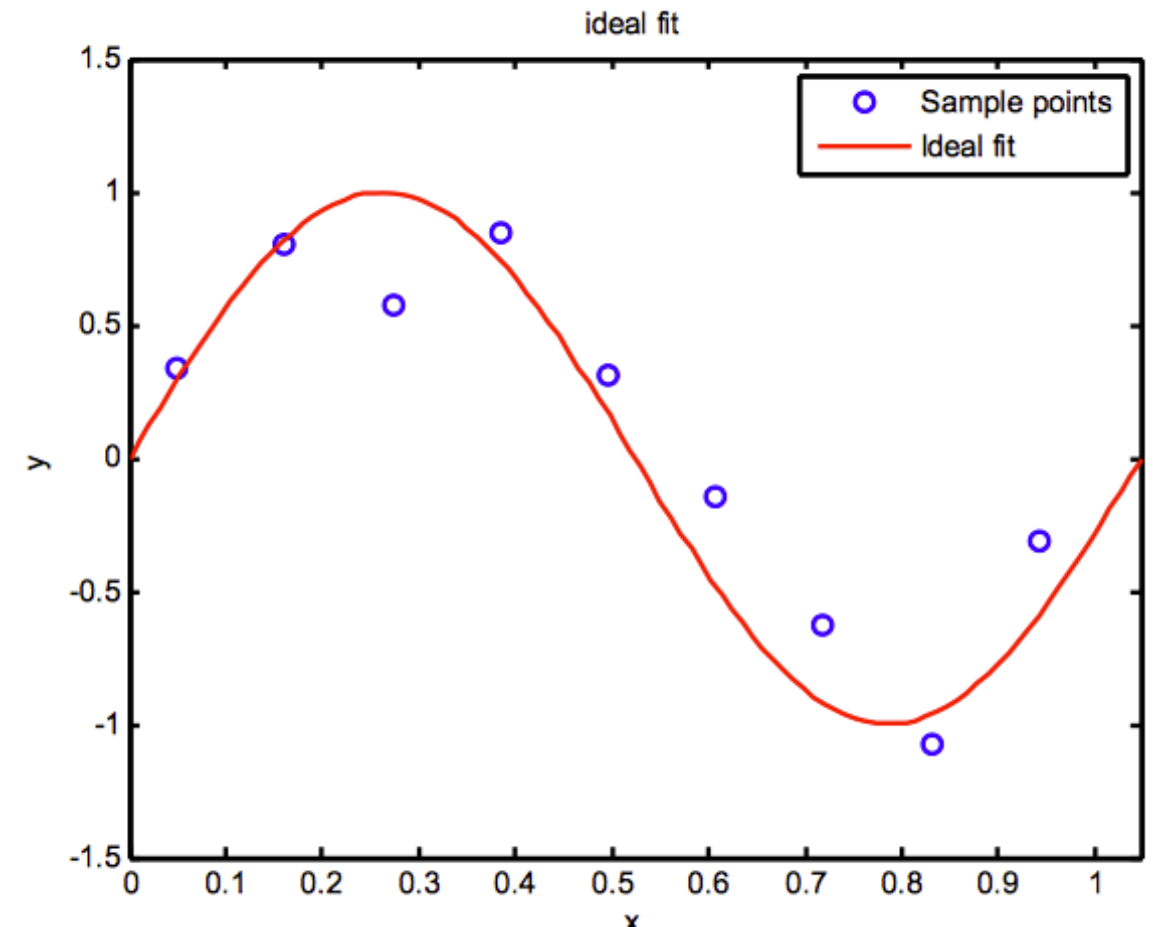
$$\frac{\partial \tilde{E}(\theta)}{\partial \theta} = -z^T (y - z\theta) + \lambda \theta$$

$$(z^T z + \lambda I) \theta = z^T y$$

$$\theta = (z^T z + \lambda I)^{-1} z^T y$$

Ridge Regression Example

- The red curve is the true function (which is not a polynomial)
- The data points are samples from the curve with added noise in y .
- There is a choice in both the degree, D , of the basis functions used, and in the strength of the regularization

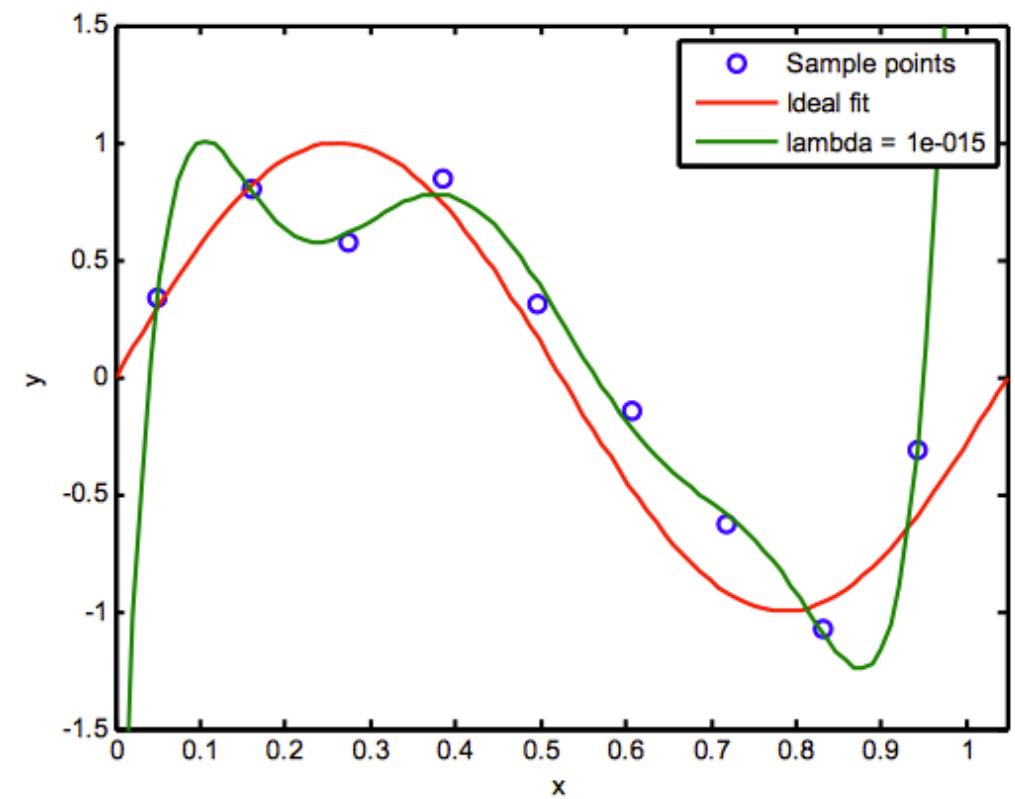
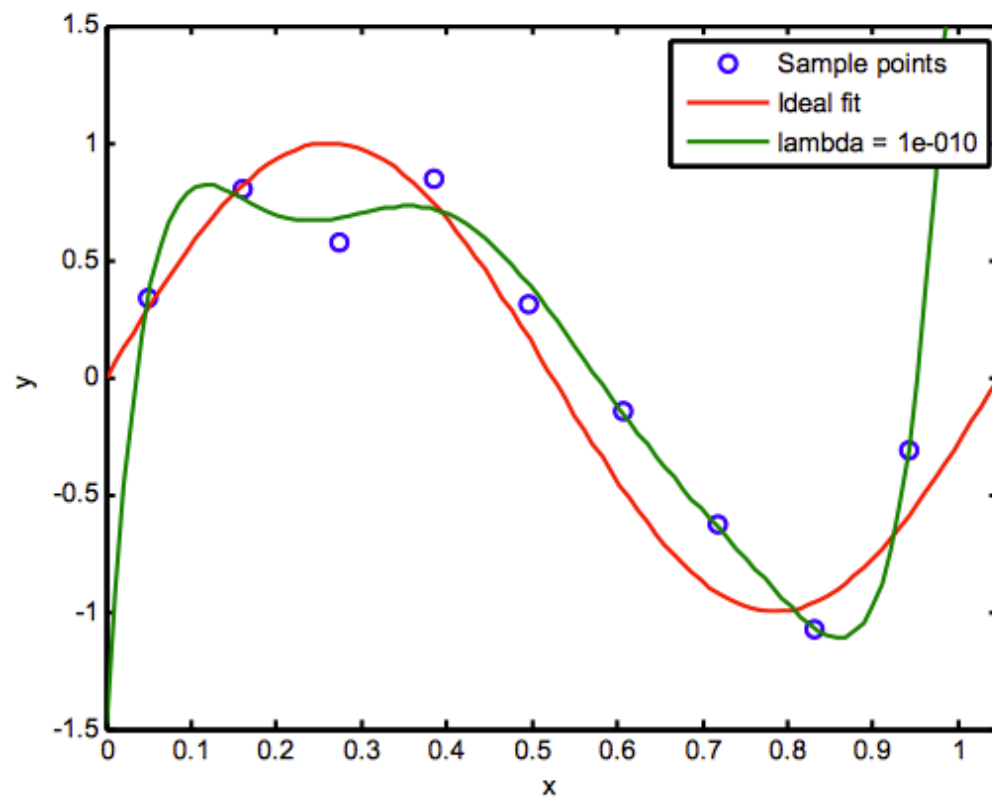
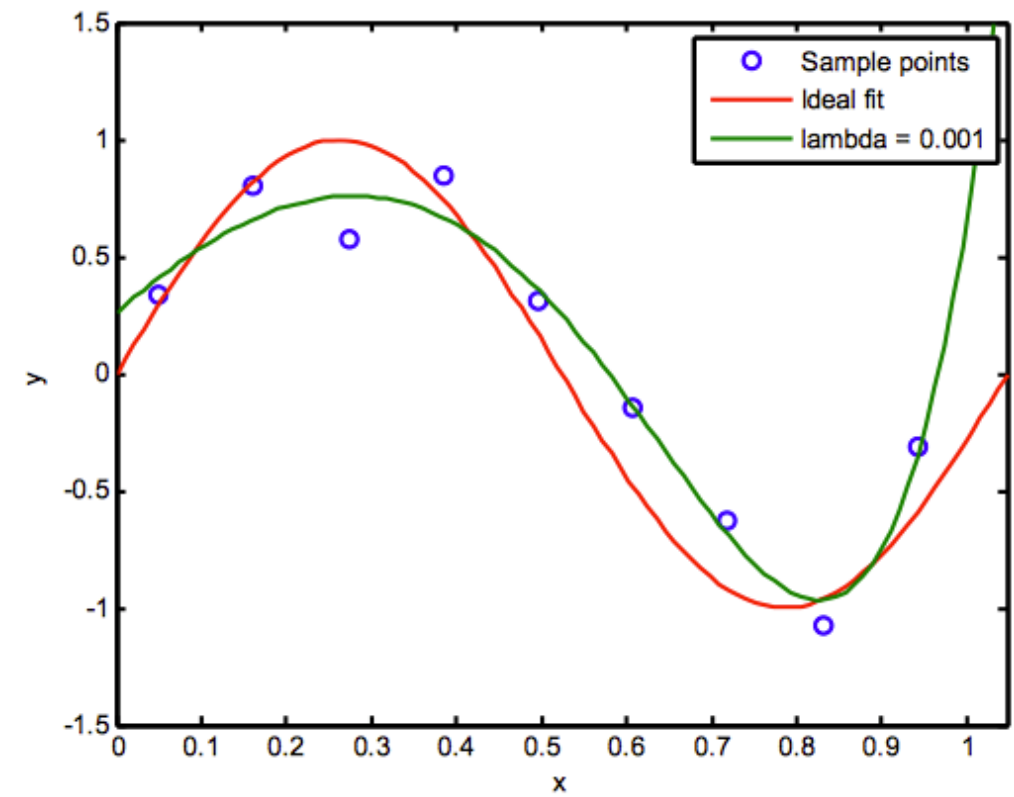
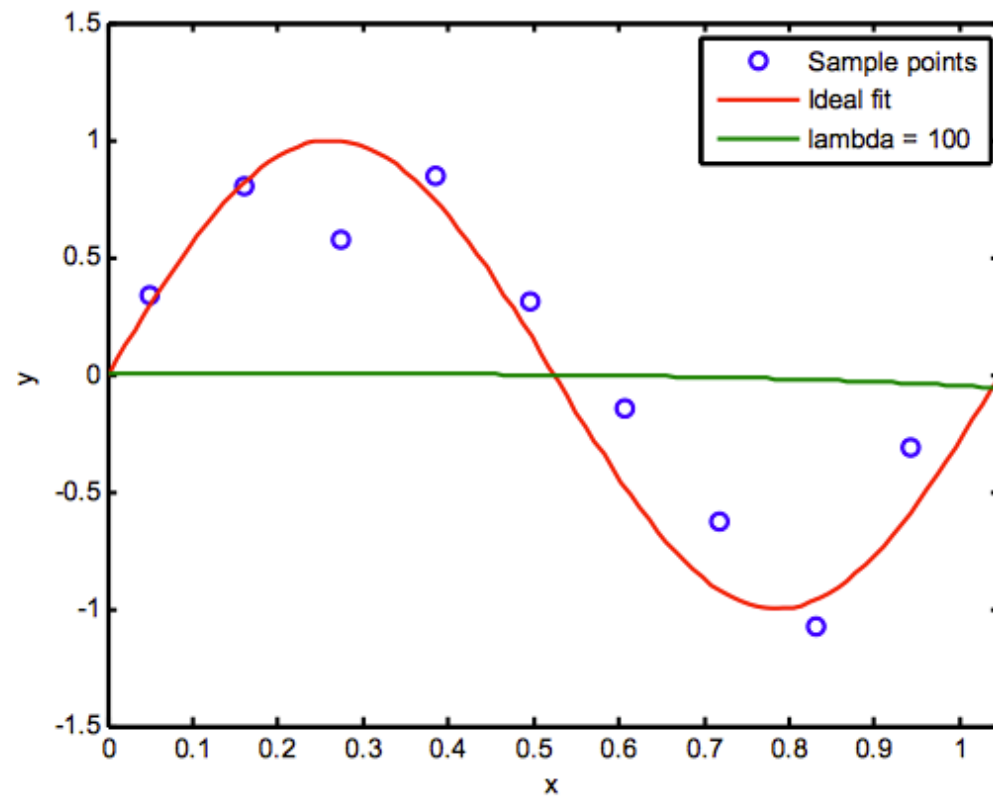


$$f(x, \theta) = z\theta$$

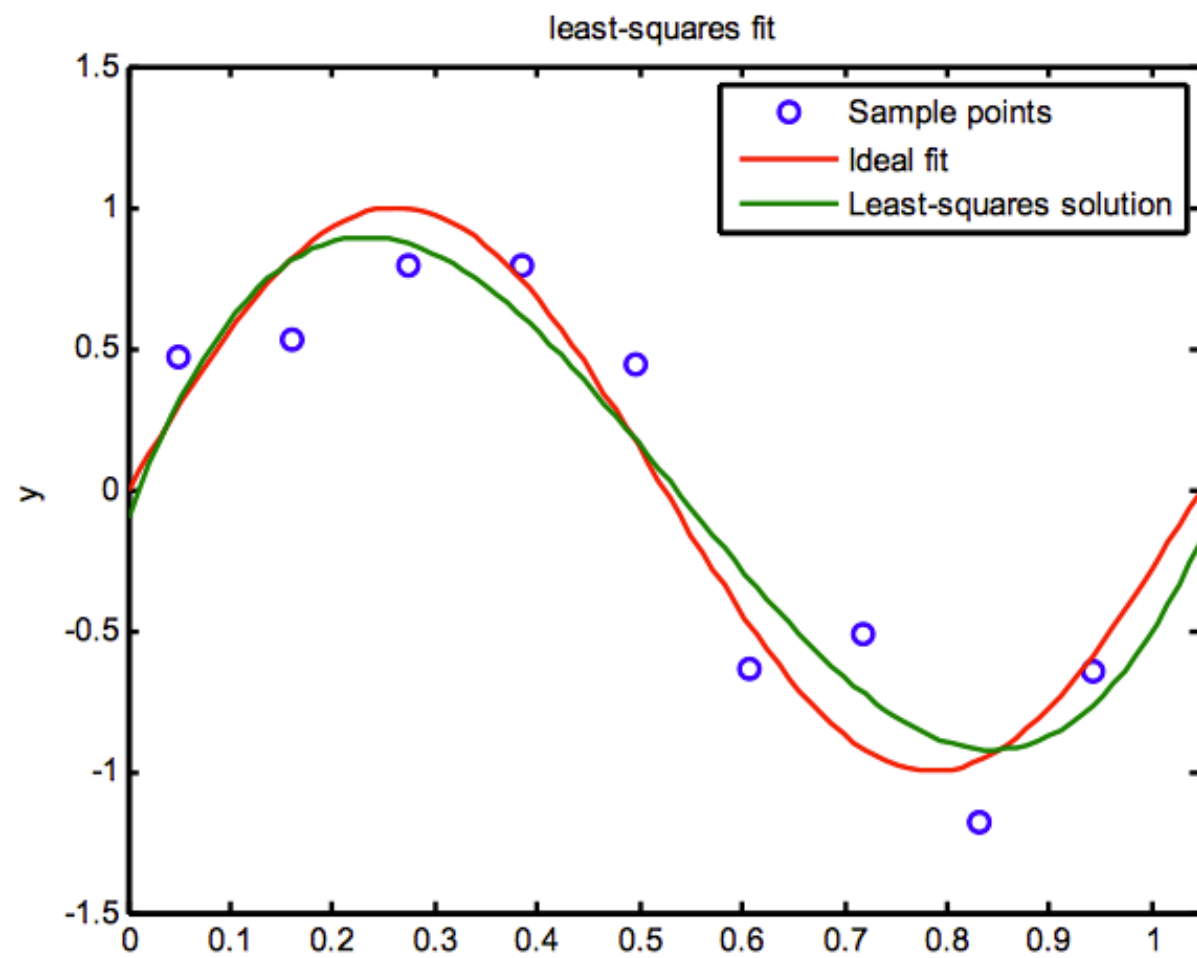
$$z: x \rightarrow z$$

$$\tilde{E}(\theta) = \frac{1}{N} \sum_{i=1}^n (y^{\{i\}} - z^{\{i\}}\theta)^2 + \lambda \|\theta\|_2^2 \quad \theta \in \mathbb{R}^{D+1}$$

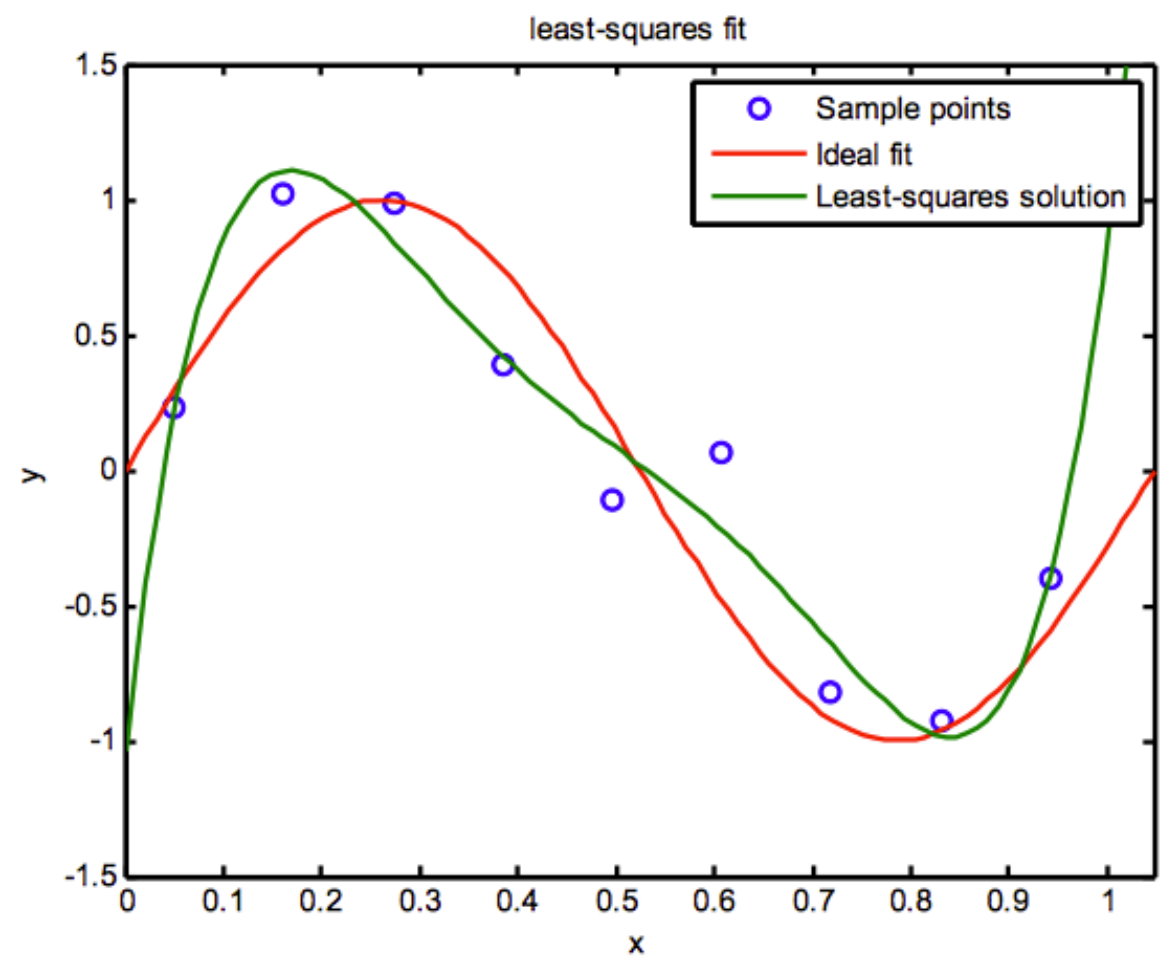
$N = 9$ samples, $D = 7$



$D = 3$



$D = 5$



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Regularized Regression

$$\tilde{E}(\theta) = \frac{1}{N} \sum_{i=1}^n (y^{\{i\}} - z^{\{i\}} \theta)^2 + \lambda \|\theta\|_2^2$$

Squared loss/Error

$$\frac{1}{N} \sum_{i=1}^n (y^{\{i\}} - z^{\{i\}} \theta)^2$$

L2 Regularizer

$$\lambda \|\theta\|_2^2$$

Now let's look at another regularization choice.

The Lasso Regularization (L1 norm) and sparsity

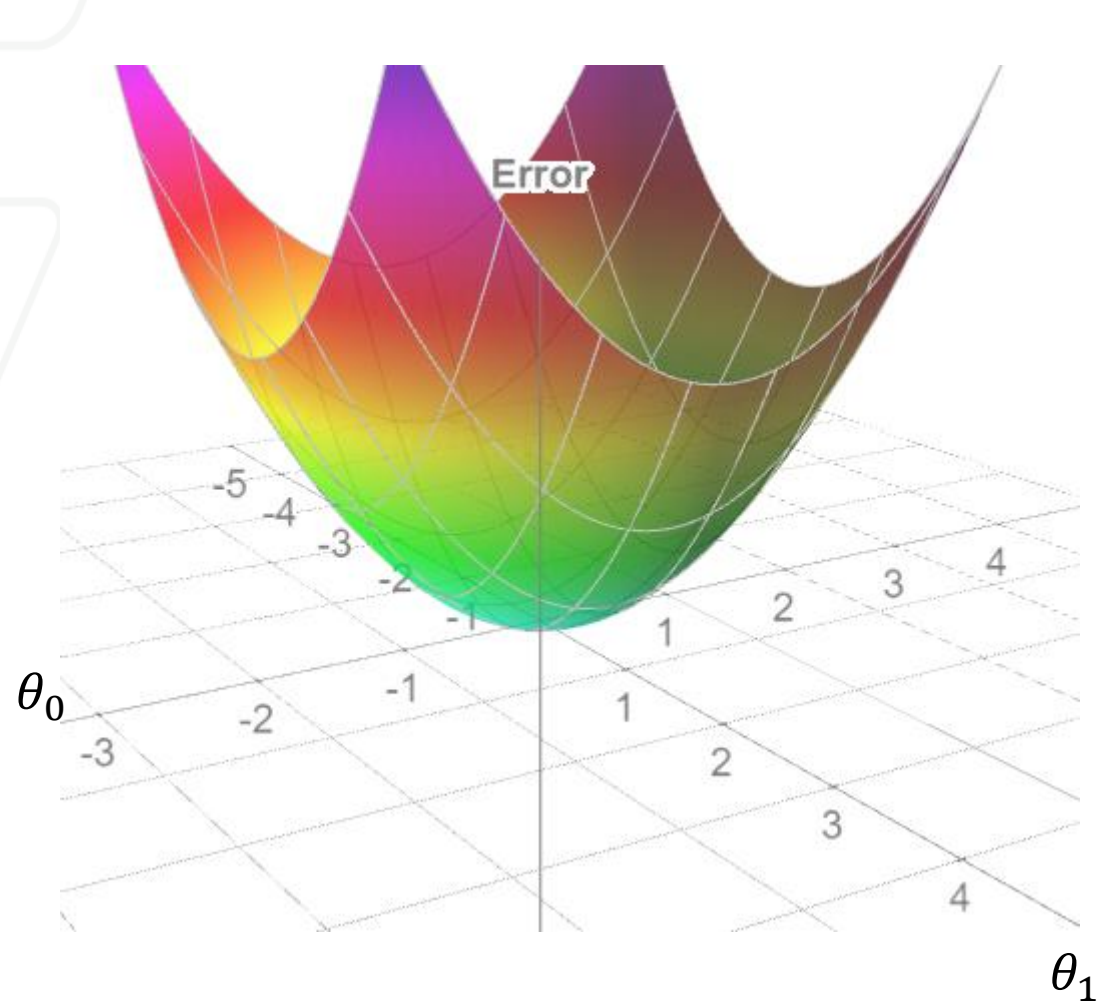
Lasso = **L**east Absolute **S**hrinkage and **S**election **O**perator

$$\tilde{E}(\theta) = \frac{1}{N} \sum_{i=1}^n (y^{\{i\}} - z^{\{i\}} \theta)^2 + \lambda \|\theta\|_1$$

L1 norm induces sparsity. This means that some of the weights become zero, and the feature contribution will be completely removed. L1 Regularizer could be used for feature selection

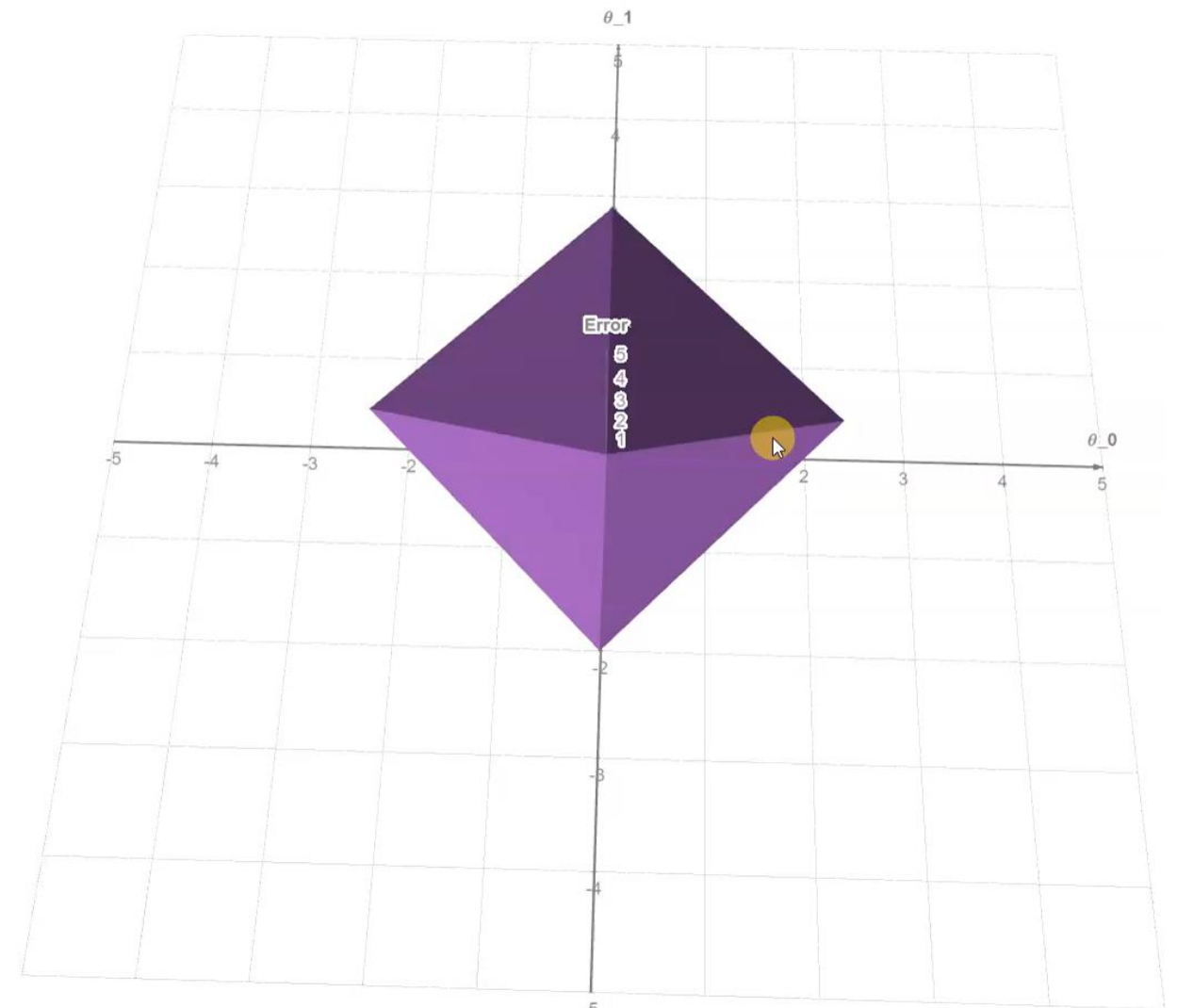
Ridge Regularizer

$$g(\theta) = \theta_0^2 + \theta_1^2 = \theta^T \theta$$



Lasso Regularizer

$$g(\theta) = \theta_0 + \theta_1 = \theta$$



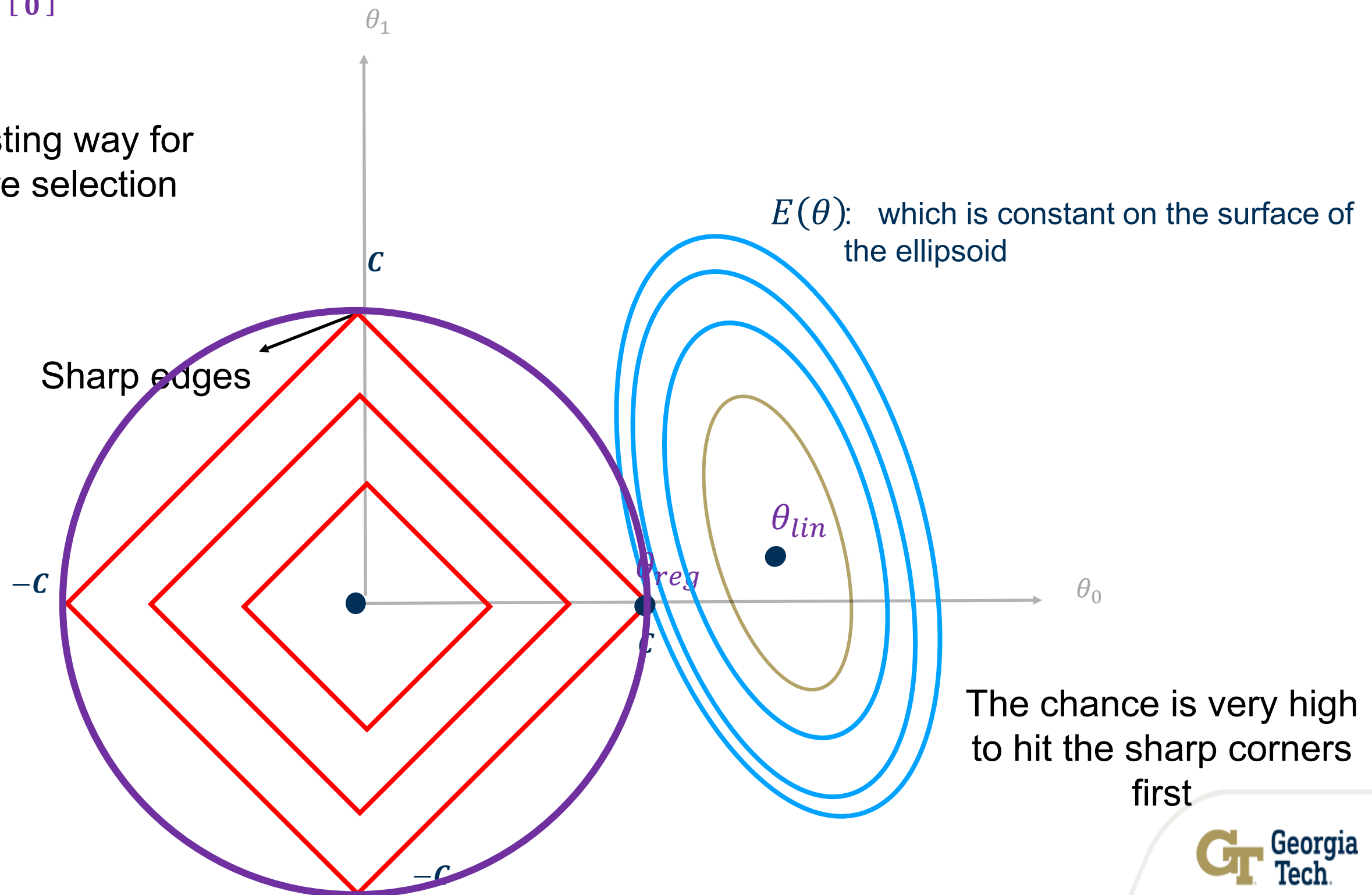
[Animation](#)

Let's say we have two parameters (θ_0 and θ_1)

$$\text{Min } E(\theta) = \frac{1}{N} (Z\theta - y)^T (Z\theta - y) + \lambda \|\theta\|_1$$

$$\theta = \begin{bmatrix} \theta_0 \\ \mathbf{0} \end{bmatrix}$$

Interesting way for feature selection



Ridge versus Lasso

Ridge

$$\begin{aligned}\tilde{E}(\theta) \\ &= \frac{1}{N} (y - z\theta)^T (y - z\theta) + \lambda \|\theta\|_2^2\end{aligned}$$

It is a convex model

Both mean squared error and L2 regularizer are differentiable.

We can get a closed form solution

Lasso

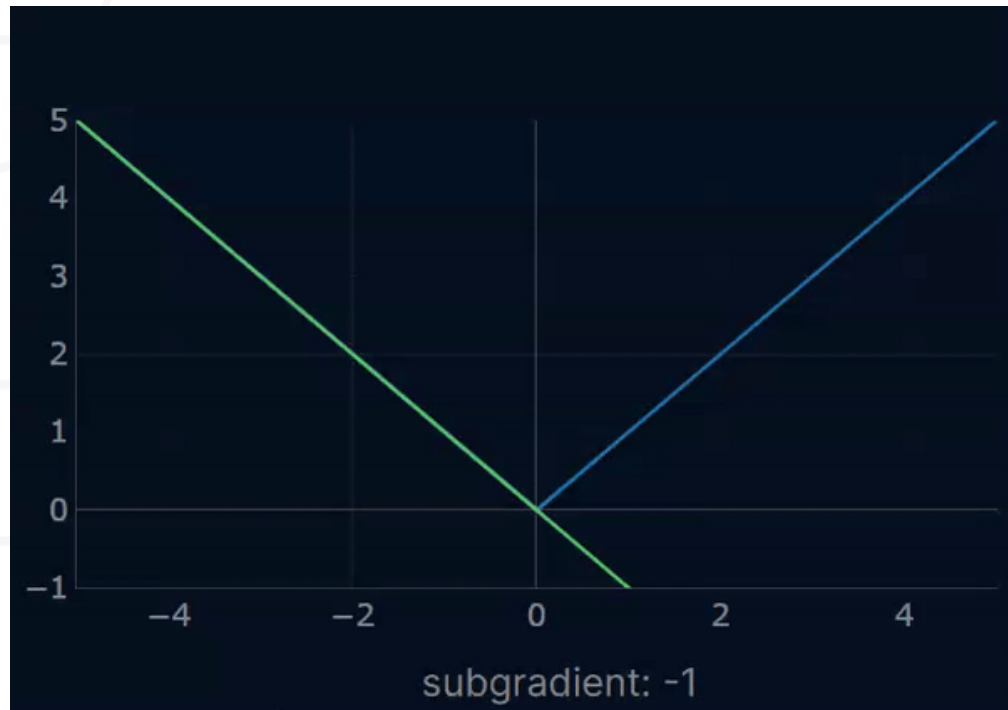
$$\begin{aligned}\tilde{E}(\theta) \\ &= \frac{1}{N} (y - z\theta)^T (y - z\theta) + \lambda \|\theta\|_1\end{aligned}$$

It is a convex model

L1 regularizer is NOT differentiable.

We can **NOT** get a closed form solution

Sub-gradient Descend in Lasso

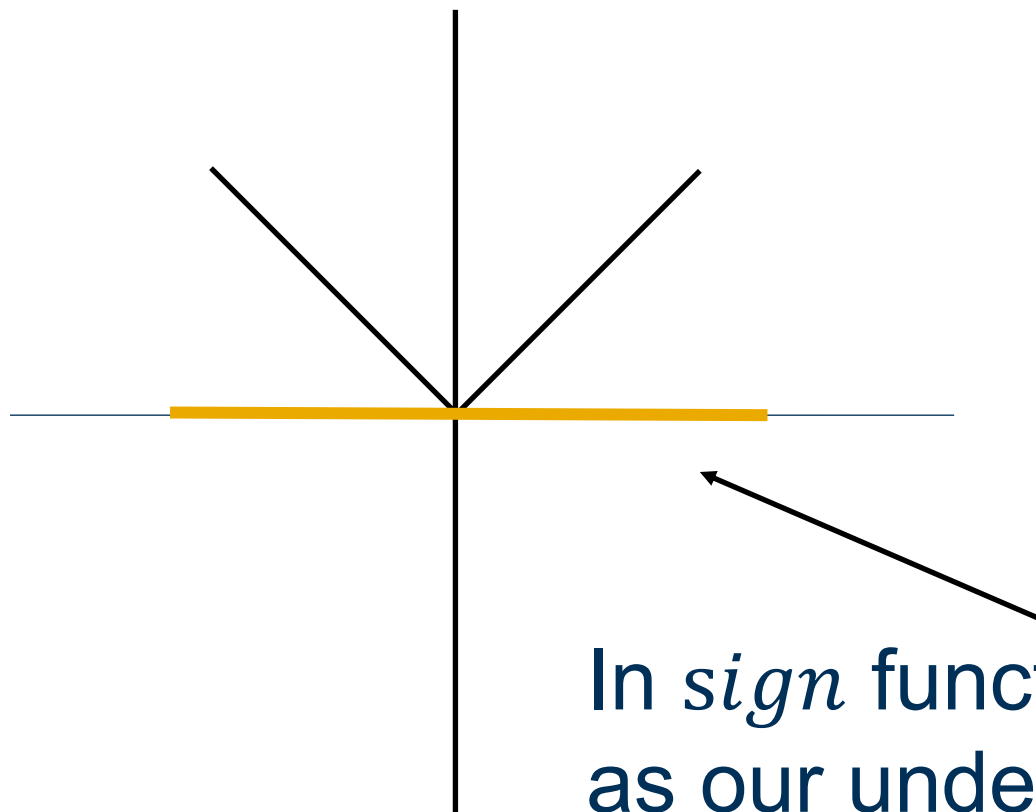


$$\tilde{E}(\theta) = \frac{1}{N} (y - z\theta)^T (y - z\theta) + \lambda \|\theta\|_1$$

$$\frac{\partial \tilde{E}(\theta)}{\partial \theta} = -z^T (y - z\theta) + \frac{\partial (\lambda \|\theta\|_1)}{\partial \theta}$$

Using Sub-gradient

$$\frac{\partial \tilde{E}(\theta)}{\partial \theta} = -z^T (y - z\theta) + \lambda \text{sign}(\theta)$$



In *sign* function, we use this sub-gradient line as our under-estimator (below our function)

Recap

- Why do we need regularization?

to prevent overfitting

$$\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_d x_d$$

- What is the root cause?

High values of θ

$$\min_{\theta} E(\theta) = \frac{1}{N} \sum_i (y_a^{(i)} - x^{(i)T} \theta)$$

- How do we regularize in ridge regression?

- Why is the penalty term chosen as $\theta^T \theta$?

$$\tilde{E}(\theta)_{\text{Regularized error}} = E(\theta) + \lambda(\theta^T \theta)_{\text{penalty term}}$$

Recap

$$\tilde{E}(\theta) = E(\theta) + \lambda(\theta^T \theta)$$
$$\boxed{\tilde{E}(\theta) = E(\theta) + \lambda \|\theta\|_2^2} \rightarrow \frac{\partial \tilde{E}}{\partial \theta} = 0$$

- New regularized objective function?

$$\frac{\partial \tilde{E}}{\partial \lambda} = 0$$

Implicit

- Why was lambda chosen as a hyperparameter to optimize the objective function?

- Small vs high values of lambda

$$\theta = (X^T X)^{-1} X^T y \leftarrow \text{closed form}$$

non regularized

$$\theta_{\text{regularized}} = (X^T X + \lambda I)^{-1} X^T y \leftarrow \text{closed form}$$

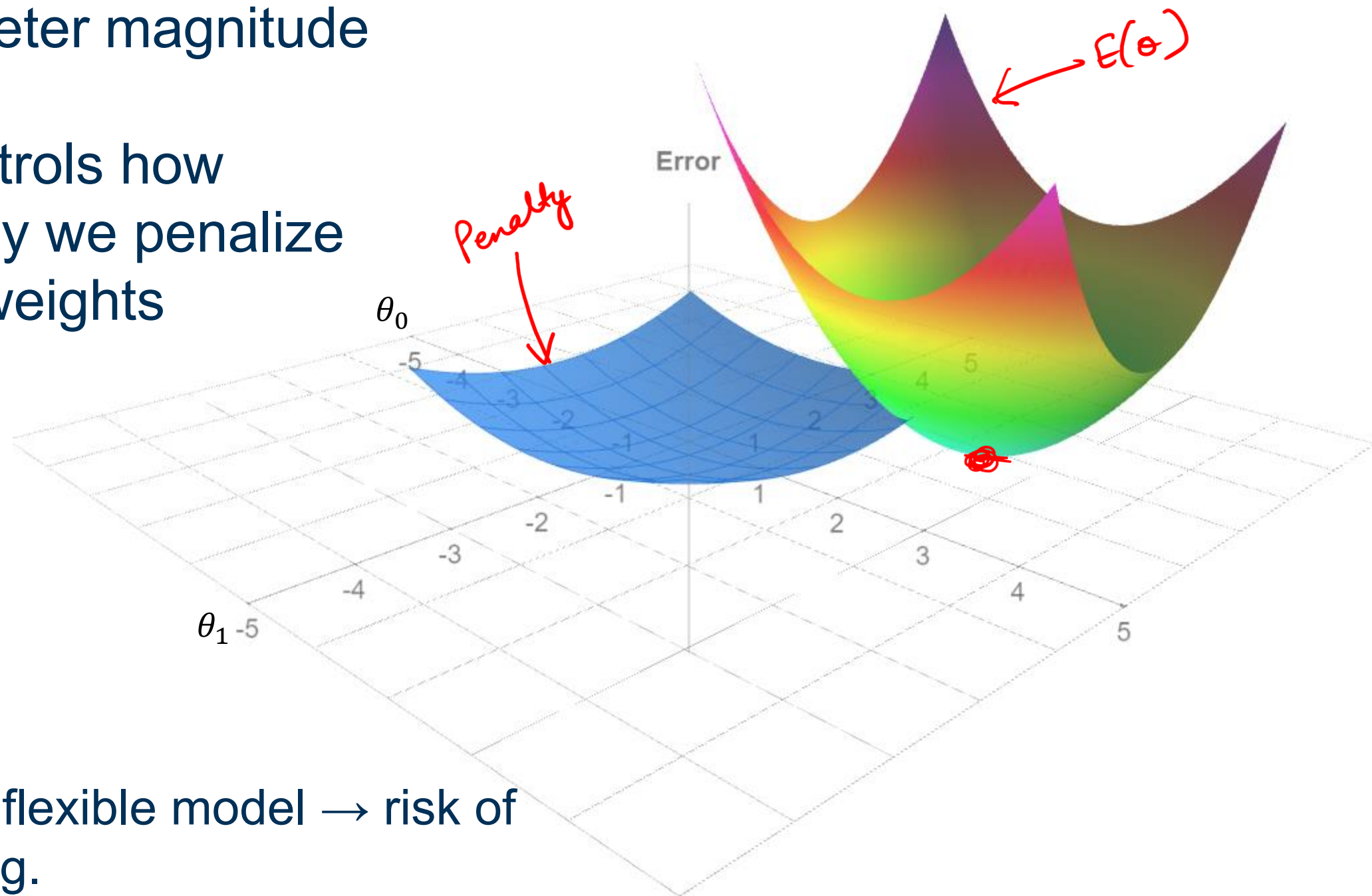
↓
Gradient Descent

The effect of low Lambda

$\frac{\lambda}{N} \theta^T \theta$: The **regularization penalty** on parameter magnitude

λ : Controls how strongly we penalize large weights

$$E(\theta) + \frac{\lambda}{N} \theta^T \theta$$

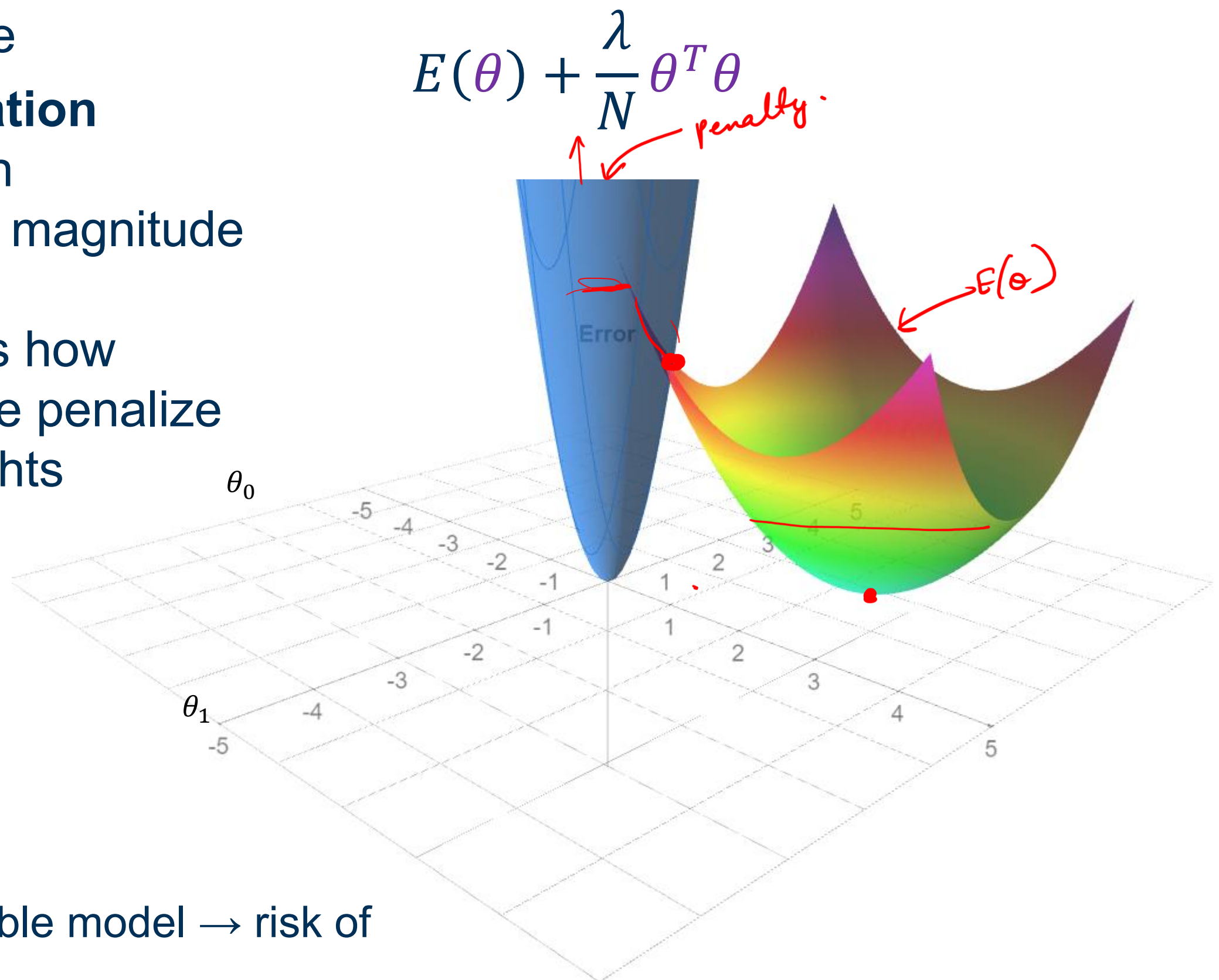


low $\lambda \rightarrow$ flexible model $\rightarrow$ risk of overfitting.

The effect of high Lambda

$\frac{\lambda}{N} \theta^T \theta$: The **regularization penalty** on parameter magnitude

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high $\lambda \rightarrow$ stable model $\rightarrow$ risk of underfitting.

Recap

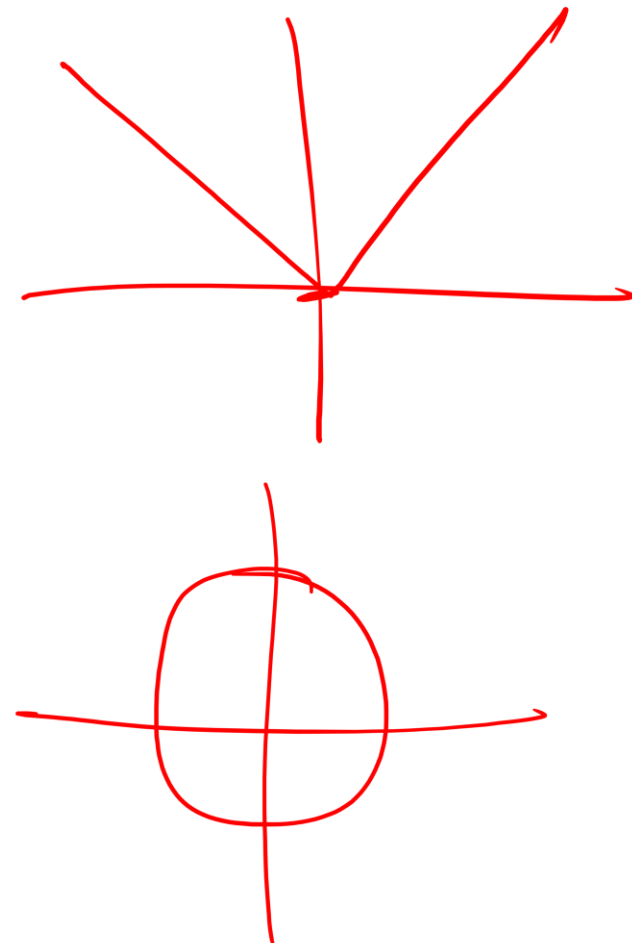
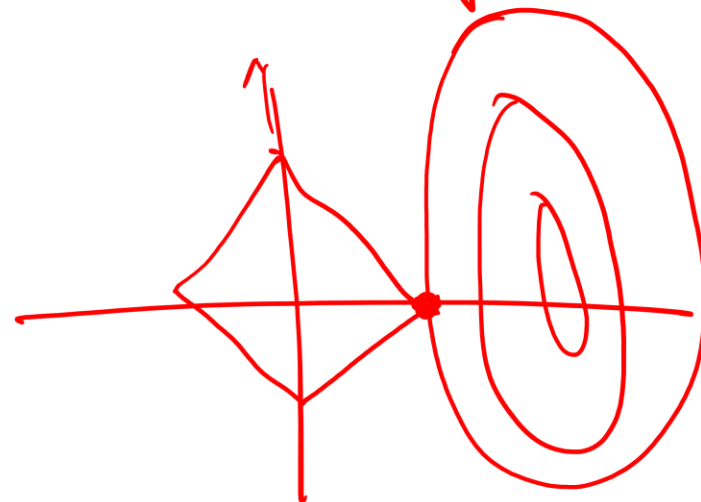
- Other forms of regularization covered in previous classes?

PCA

- L1 vs L2 Regression?

$$+ \lambda \| \theta \|_2^2$$
$$+ \lambda \| \theta \|_1$$

helps in dimensionality reduction



$$\{i\} y_a - E[\{i\} y_p]$$

Regulariza tion	Bias	Variance	Total Error	Stability
✗ Without	<u>Low</u>	<u>High</u>	High	Overfits
✓ With	<u>Slightly higher</u>	<u>Much lower</u>	<u>Lower</u>	Generalize s better

$\lambda \leftarrow$ hyperparameter



**THE BEST WAY TO
EXPLAIN OVERFITTING**

Midsemester Feedback

- Overall very positive
- Lecture Effectiveness: 4.12/5. std. dev: 0.6/5
- Quiz and Project Effectiveness: 4.18 and 4.16, respectively.
- Main Good Feedback about lectures: Engaging, well-explained
- Main Criticism Points:
 - Sometimes the pace is fast and not well balanced across lectures
 - GMM could be better
 - Sometimes I move to next slide after writing on the slide
- Midterm main criticism point: the practice exam could have been more similar
- Overall, most important feedback for me:
 - 53% positive sentiment
 - 6% negative
 - 41% neutral

Outline

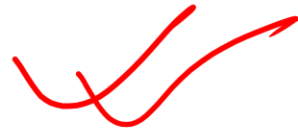
- Overfitting and regularized learning
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Train

Val

Test



Fix the
optimum
value of
hyperparameters

NOT GOOD

Leave-One-Out Cross Validation

Choose λ with lowest average test error

$n=100$

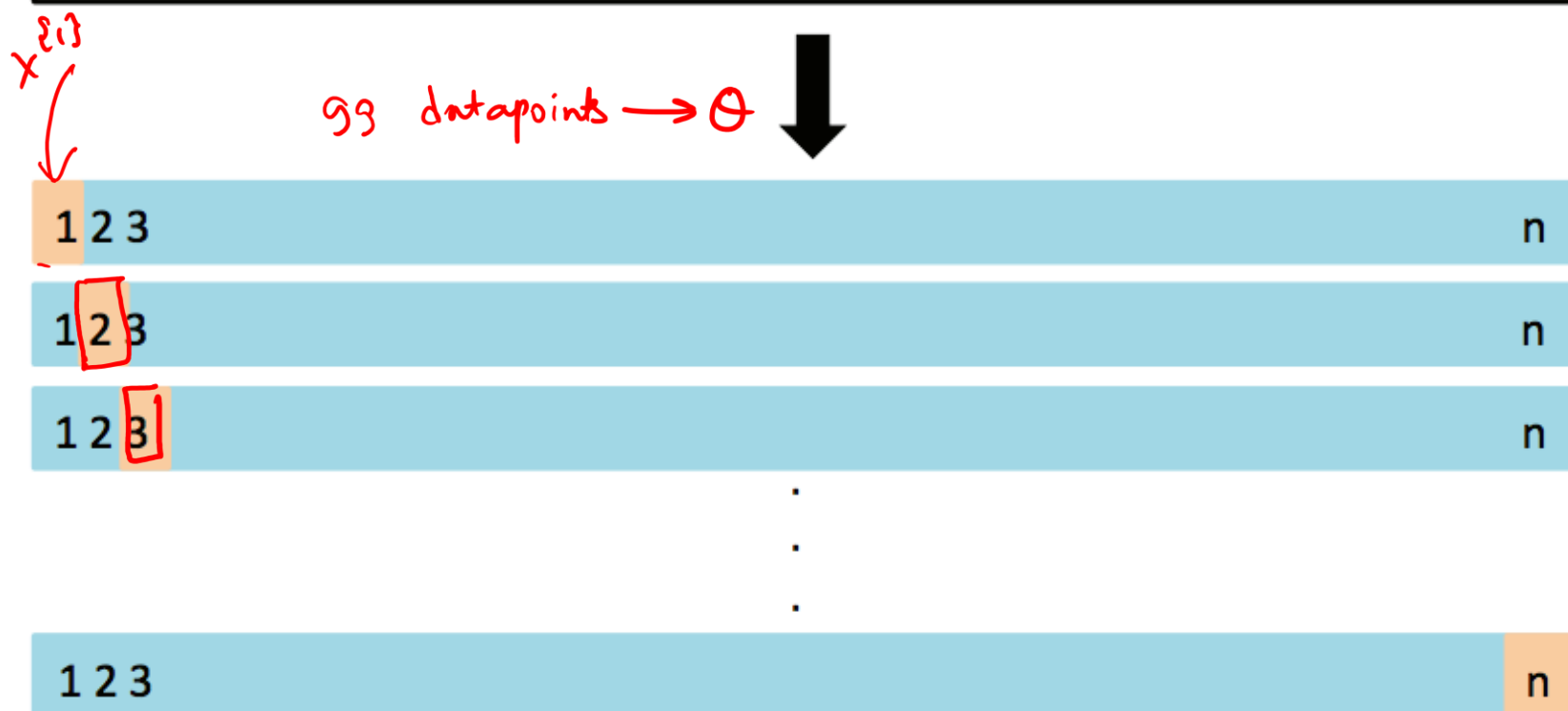
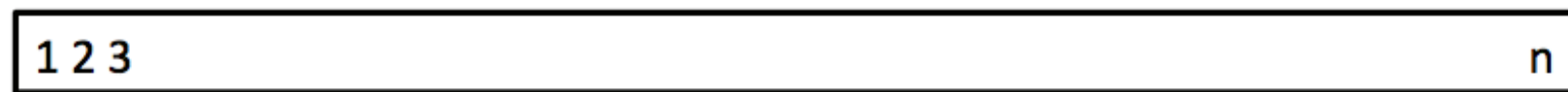
$\lambda = 0.0001$
 $\lambda = 0.001$
 $\lambda = 0.01$
 $\lambda = 0.1$
 $\lambda = 1$

For every $i = 1, \dots, n$:

- ▶ train the model on every point except i ,
- ▶ compute the test error on the held out point.

Average the test errors.

$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i^{(-i)})^2$$



$$E_1 = y^{(1)} - x^{(1)} \cdot \theta$$

E_2

E_{100}

$$\frac{1}{n} \sum E_i \leftarrow \lambda$$

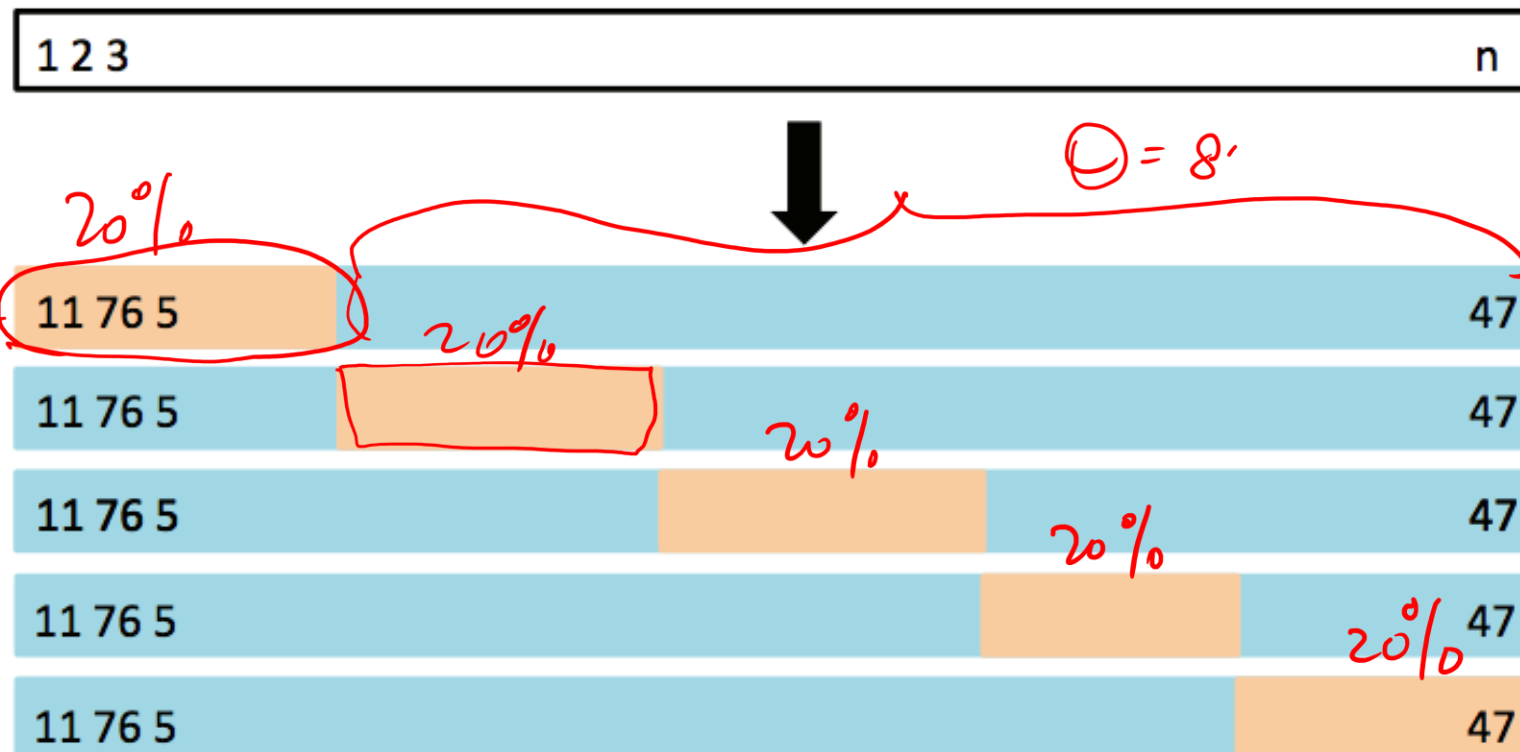
K-Fold Cross Validation

Split the data into k subsets or *folds*.

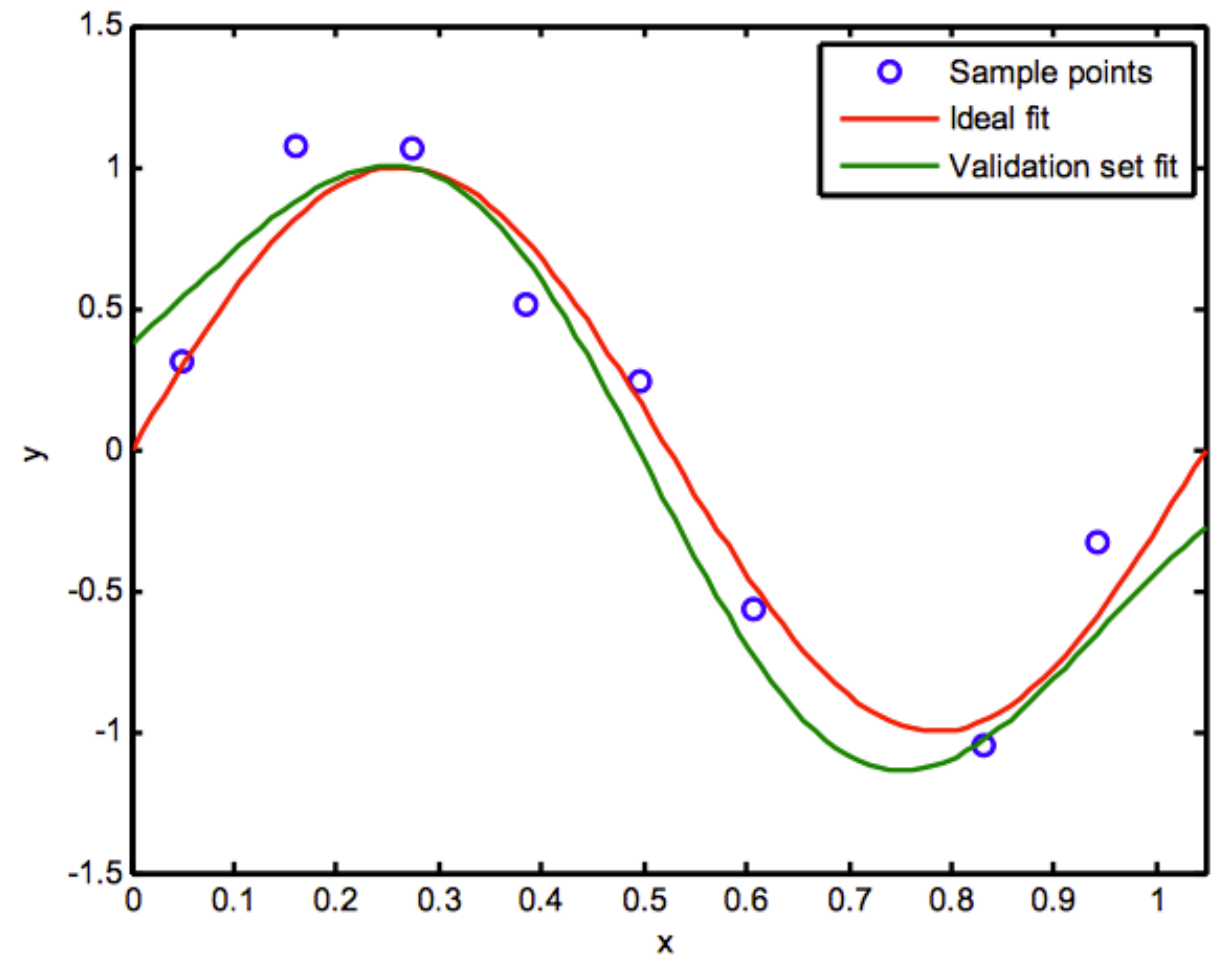
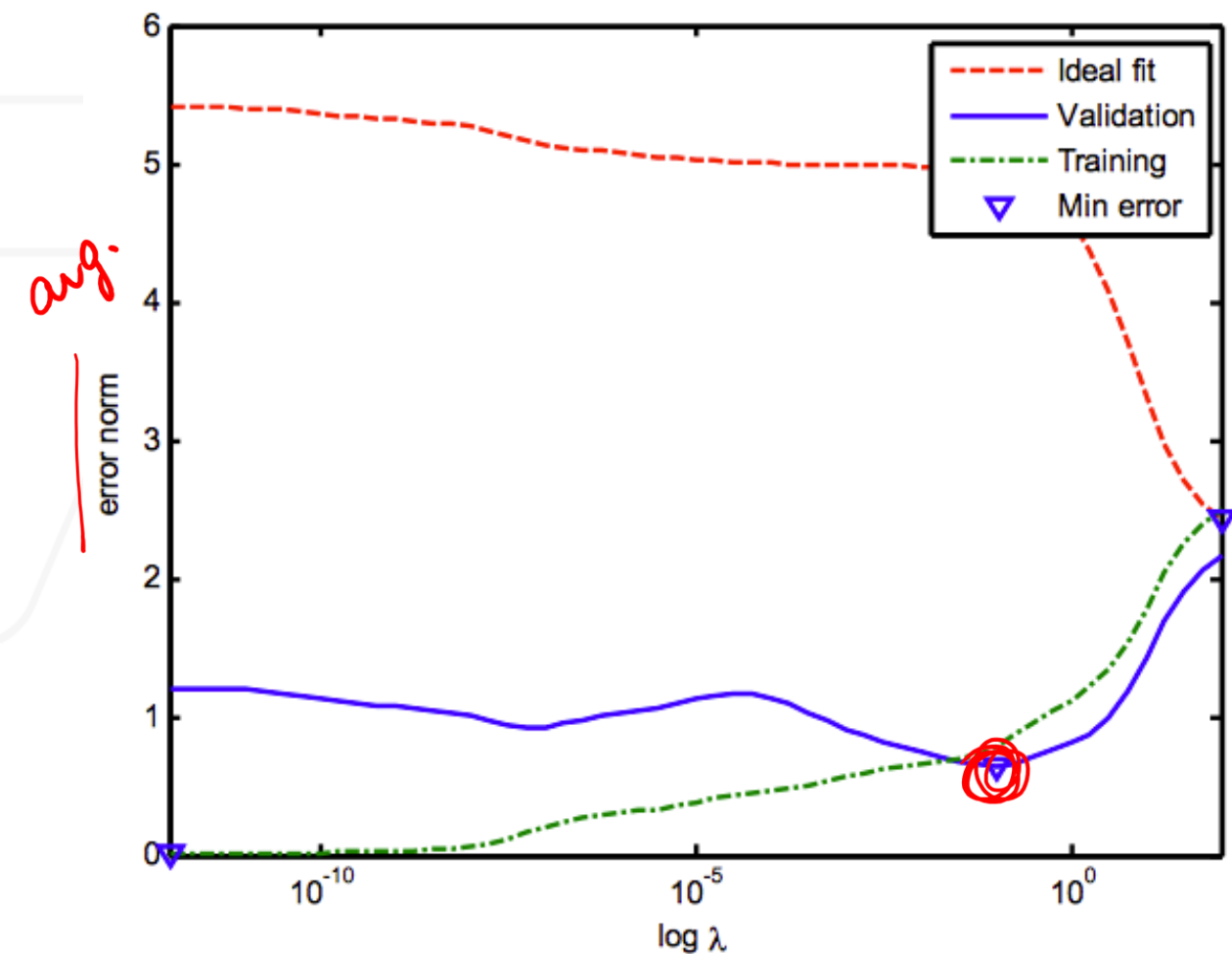
For every $i = 1, \dots, k$: ⁵ disjoint subsets

- ▶ train the model on every fold except the i th fold,
- ▶ compute the test error on the i th fold.

Average the test errors.



Choosing λ Using Validation Dataset



Pick up the lambda with the lowest mean value of mse calculated by Cross Validation approach

Take-Home Messages

- What is overfitting
- What is regularization
- How does Ridge regression work
- Sparsity properties of Lasso regression
- How to choose the regularization coefficient λ

Quick Knowledge Check

1. Which of the following is the *main reason* for overfitting in regression? A. Too few features. B. Too small a training dataset. C. Large model weights. D. High bias
2. When λ (lambda) is **very small**, what happens to the model? A. High bias, low variance. B. Low bias, high variance. C. Both bias and variance decrease. D. Both bias and variance increase
3. Which regularization technique results in **sparser** models? A. Lasso (L1). B. Ridge (L2). C. Both equally. D. Neither
4. In **k-fold cross-validation**, what does each fold provide? A. A new model parameter. B. A separate test set. C. A different λ value. D. A validation error estimate
5. Fill in the blank for ridge regression:

$$E(\theta) \cancel{J(\theta; \lambda)} = \frac{1}{2n} \|y - X\theta\|^2 + \frac{\lambda}{2} \underline{\quad}$$

- A. $\|\theta\|$ B. $X^T \theta$ C. $\|\theta\|^2$ D. $|\theta|$